

# VP Chemistry & Electromagnetism

Derived from a Single Anchor on the Jamming-Lattice Substrate — Foundations, Optics,  
and Applications

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## Abstract

This volume consolidates the Volume-Particle (VP) account of **electromagnetism, chemistry, and their engineering applications** into a single reproducible whitepaper. Everything is derived from one observational anchor — the electron, treated as a rotational dent in an infinitely rigid, fully packing “jamming” lattice — using only  $\pi$ -geometry and the electromagnetic inverse-square law. It is a companion to the VP Theory (physics) whitepaper, from which it takes the fundamental electromagnetic mechanism (§10, §14) as input.

Three results anchor the account. First, **light is not assumed but emerges**: at the isostatic point of the jammed lattice the relaxed shear modulus vanishes while the bulk modulus stays finite, so a single longitudinal wave survives at  $c^2 = B/\rho$ ; a deterministic lattice simulation reproduces this to 0.06%. Second, on that substrate charge is synchronized rotation, the Coulomb  $1/r^2$  follows from three-dimensional flux conservation, the  $E/B$  geometry and polarization are rotation, and the same wave spans conduction and radiation by a single angle  $\sin \chi = \lambda/(mD)$ ; **and the dynamic field equations, the Poynting energy flow, and a strict light-cone causality are supplied in longitudinal form** (Chapter EM, §EM.12, from the AQD axiomatic field dynamics). Third, chemistry follows by sphere geometry and rotation — bonding angles  $\arccos(-1/3)$ , the crystal-field ratio 4/9, the close-packing fraction 0.7405, the nuclear magic skeleton, thermodynamics, equilibrium, kinetics, solution and electrochemistry — and the engineering applications (catalysis, fertilizer, water, energy storage, new materials, and **CO<sub>2</sub> reduction**) are re-grounded on the **verified  $d$ -band energy descriptor**, not on the refuted electron-amplitude variable.

Every forced result has zero free parameters and is therefore falsifiable; measured constants are labelled as calibration inputs and never tuned per result; open problems — most fundamentally the electromagnetic coupling  $\alpha_{\text{em}}$  itself — are admitted rather than disguised. All numbers are reproduced by deterministic, standard-library Python modules (a 38-module chemistry/EM core with a 176-row case ledger, plus a 9-module / 46-row applications package) with verification harnesses (core 38/38 pass; applications 9/9 pass) and numeric-

drift gates that check every displayed number against canonical recomputation and module output (both pass, zero drift).

**Keywords:** Volume-Particle theory, jamming lattice, light emergence,  $c^2 = B/\rho$ , electromagnetism, Poynting, causality, Coulomb  $1/r^2$ , propagation angle, VSEPR, crystal-field 4/9,  $d$ -band catalysis,  $\text{CO}_2$  reduction, reproducibility, falsifiability.

## Introduction

### The program and this consolidation

The VP framework posits a vacuum built from infinitely rigid, fully packing particles — a jammed lattice. From this single substrate the physics whitepaper derives the speed of light, the proton radius, and a ladder of dimensionless constants. This whitepaper extends the program into **electromagnetism, chemistry, and applications**, asking how much of each follows with *no adjustable parameters* once light has emerged. This consolidated edition gathers what were previously separate chapter files (EM, CH, CM, CC, CT) and the applications package (CA) into one volume, and incorporates two completions made after the first release: the **dynamic-field / Poynting / causality** addendum (§EM.12) and the  **$\text{CO}_2$  electrocatalytic reduction** module (§CA, re-grounded on the  $d$ -band descriptor).

### Why light first

The organizing principle is sequencing: **light must emerge before anything else follows**. Once the lattice supports a single longitudinal wave at  $c^2 = B/\rho$  with linear dispersion  $\omega = cq$ , a wavelength exists; once a wavelength exists, the propagation angle, the quantum diameter, refraction, and color follow; and once charge is identified as synchronized rotation, the Coulomb law and the entire chemical edifice follow. Chapter EM begins with the emergence of light and develops electromagnetism (now including its dynamic and causal structure); the chemistry chapters build on that foundation; the applications chapters close the arc on verified physics.

### How claims are graded (the No-Tuning discipline)

- **[F] forced** results are dimensionless geometry or integers with nothing to adjust (the tetrahedral angle  $\arccos(-1/3)$ , the ratio 4/9, the packing fraction  $\pi/(3\sqrt{2})$ ). They cannot be tuned, so they are *refuted* if measurement disagrees.
- **[CAL] calibration inputs** are measured constants (the fine-structure coupling, spectroscopic constants, standard potentials,  $d$ -band centers) fixed once and used consistently; quantities *derived* from them and then matching measurement are predictions, not fits.
- **[V] simulation-measured**, **[H] hypothesis**, and **[O] open** label what a deterministic simulation shows, what remains a working hypothesis, and what is honestly unsolved.

A central methodological result of this program is **negative**: the early “electron amplitude (fm)” mechanism for catalysis, bond-angle deviations, magnetic desalination, and black-copper power generation was tested and **refuted**, and is replaced — where a working mechanism exists — by the correct variable (the  $d$ -band *energy*, equilibrium thermodynamics, real magnetostatics, Carnot-bounded storage). Chapter CT records this refutation in full; the applications chapter is built on the replacement. Letting the refuted mechanism go is what makes the surviving results trustworthy to build on.

### Reproducibility as a first-class requirement

Every numeric claim is produced by a deterministic module using only the Python standard library; determinism is verified by re-running and comparing output hashes. Verification harnesses check execution, determinism, dependency, and case-ledger integrity (chemistry/EM core: 38/38; applications: 9/9), and numeric-drift gates

check that every displayed number matches both the canonical closed-form recomputation and the module output (both pass, zero drift). Nothing here is asserted that the reader cannot re-run.

## Boundary with the physics whitepaper

The *fundamental* electromagnetic mechanism — light as a rotational transverse wave, the single surviving longitudinal speed, and the  $1/r^2$  Green function — is established in the physics whitepaper (§10, §14). This document consumes those results and develops the chemical, optical, and dynamical consequences. The electromagnetic coupling strength  $\alpha_{\text{em}}$  is **not derived** in this framework (physics §14.5); it is an honest open item, carried as [CAL].

# Electromagnetism in the Jamming Vacuum: From the Quantum Lattice to Light

**VP Chemistry & Electromagnetism Whitepaper — Chapter EM** DOI (reserved): 10.5281/zenodo.20680541 · Companion to VP Theory (physics) §10, §14, §15 Status: each result is graded **[F]** forced (zero free parameters) · **[CAL]** calibration input (measured constant, used consistently) · **[V]** simulation-measured · **[H]** hypothesis · **[O]** open. Every numeric claim is reproduced by a deterministic, standard-library module ( $2 \times \text{sha256}$  identical); module names are given inline.

One medium, one motion seen two ways. Charge is the bookkeeping of a synchronized rotation; light is the transverse swing of that rotation propagating on the single surviving longitudinal wave. Electricity, a radio wave, and a  $\gamma$ -ray differ only by an angle.

## EM.0 Governance (inherited from VP Theory §1)

This chapter inherits the No-Tuning / LOCK / Gate discipline of the physics whitepaper. **The fundamental electromagnetic mechanism is supplied by physics §10 (light, clock-free  $c$ ) and §14 (force, the  $1/r^2$  Green function). This chapter does not re-derive it; it takes those results as input and develops the chemical and optical consequences.** Where a quantity is a measured constant (notably the fine-structure coupling  $\alpha_{\text{em}}$ ), it is marked **[CAL]** and is never tuned to fit a downstream result.

## EM.1 Genesis: charge as synchronized rotation **[H]**→**[V]**

A quantum that receives energy can only store it as rotation, so energy input forces rotation. This is grounded in established physics: photon–photon collisions create matter ( $\gamma\gamma \rightarrow e^+e^-$ ;  $E = mc^2$ ), and photons carry helicity  $\pm 1$  — circular polarization is a *rotating field*. **Electric charge is the bookkeeping of one synchronized rotation** of the quanta; the sign  $\pm$  is the handedness (CW/CCW) of the twist of a charge’s rotation relative to the surrounding common axis. The elementary charge is one boundary-crossing rotation unit (universal), so the EM coupling carries no nucleon-structure factor (contrast the mass/gravity sector, where  $m_p/m_e = 6\pi^5$  and the 82+7 core appear).

## EM.2 Light emergence — the keystone **[F]**/**[V]**

Light is not assumed; it *emerges* from the jammed lattice. The chain (physics §SP):

1. **[F, axiom]** Infinitely rigid, fully packing quanta  $\Rightarrow$  the vacuum is a jammed lattice.
2. **[H]**→**[V]** Rotation (temperature) drives the contact number  $z$  to the isostatic threshold  $z = 2d = 6$  (self-organized criticality).
3. **[V]** At the isostatic point the shear reserve is zero, so the relaxed shear modulus vanishes while the bulk modulus stays finite:

$$G_{\text{relaxed}} \rightarrow 0, \quad B = \text{finite} \Rightarrow c^2 = \frac{B}{\rho} = K.$$

The transverse acoustic wave dies; **one longitudinal speed survives.**

4. **[F]** A single speed gives linear dispersion and hence a wavelength:

$$\omega = cq \Rightarrow \lambda = c/\nu.$$

**Direct demonstration (module `vp_light_emergence.py`).** A 1-D chain of quanta,  $m\ddot{u}_n = k(u_{n+1} - 2u_n + u_{n-1})$ , is given a right-moving Gaussian packet and integrated by leapfrog (no randomness). The energy centroid propagates at  $c_{\text{measured}} = 0.99940$  versus the closed form  $c_{\text{theory}} = a\sqrt{k/m} = 1$  — a **0.06 %** agreement. The exact dispersion  $\omega(q) = 2\sqrt{k/m}|\sin(qa/2)|$  tends to  $\omega = cq$  as  $q \rightarrow 0$  ( $v_{ph}/c \rightarrow 0.99999$ ), and the identity  $c^2 = B/\rho$  is verified to machine precision. A 2-D triangular lattice confirms that as the shear reserve  $\rightarrow 0$  the transverse speed  $c_T \rightarrow 0$  while the longitudinal  $c_L$  survives — **light is the unique surviving longitudinal wave of the jammed lattice.**

**Quantum diameter [F].** One  $2\pi$  phase winding of the carrier sets the rotational circulation length

$$D = 2\pi a_{\text{phys}} = \frac{2\pi\lambda}{A} = 2\lambda_{C,e} = 6\pi^6 r_p, \quad r_p = \frac{2}{\pi}\lambda_{C,p},$$

where  $r_p$  is the globally stable fixed point of the stiffness-shell balance  $\alpha x^{-5} = x^{-4} \Rightarrow x^* = 2/\pi$  (physics §SP S4). Numerically  $D = 4.8526$  pm. The module verifies  $D = 6\pi^6 r_p$  to within **+18.8 ppm**, which is exactly the headline residual  $m_p/m_e = 6\pi^5$  (−19 ppm): the small mismatch is the *known residual of the theory*, not an error.

### EM.3 The Coulomb sector: why $1/r^2$ , why long-range, how strong

**Why a force exists [F].** Two sources' fields superpose; the stored strain energy depends on separation and the medium relaxes it, giving  $E_{\text{int}} = k q_1 q_2 / r$ . Like sources repel, opposite attract (a positive-definite strain structure).

**Why  $1/r^2$  [F].** A fixed source flux is diluted over the  $4\pi r^2$  shell (Gauss = Poisson Green's function). In  $d$  dimensions the field falls as  $r^{-(d-1)}$ , so  $d = 3$  gives  $1/r^2$ . Module `vp_electromagnetism.py` measures the exponent directly:  $d = 1, 2, 3, 4 \rightarrow 0, -1, -2, -3$ , confirming  $E \propto 1/r^2$  in three dimensions as the geometric consequence of flux conservation.

**Why long-range / unscreened [H].** Synchronization spontaneously breaks a *global*  $U(1)$  (the common phase), giving a massless Goldstone mode and hence infinite range. (Were the  $U(1)$  gauged, its breaking would make the mediator massive and short-ranged; long-range EM requires the broken symmetry to be global — a stated departure.)

**Coupling magnitude [CAL].** The elementary charge is one universal rotation unit, so

$$K_C = k_e e^2 = \alpha_{\text{em}} \hbar c, \quad k_e = \frac{\alpha_{\text{em}} \hbar c}{e^2}.$$

The module reproduces the Coulomb constant  $k_e = 8.987552 \times 10^9$  to  $3 \times 10^{-10}\%$ .  $\alpha_{\text{em}}$  **is taken from measurement; it is not derived here** — an honest [CAL] item.

### EM.4 The E/B geometry: longitudinal $E$ , transverse $B$ [F]

The same rotating medium supplies the vector sector. The organizing statement: **the electric field is the force directed along the light (propagation) direction; the magnetic field is the force directed at  $90^\circ$  to it.** Both originate from one fact — a charge's rotational output is *twisted* relative to the surrounding synchronized quanta because its axis differs. A charge with axis aligned to the common axis emits a pure radial (Coulomb) field and no magnetic field. A moving charge tilts its axis by an amount  $\propto v/c$ ; the propagation of that twisted region, transverse to the light direction, is the magnetic field:

$$|\mathbf{B}| = \frac{v}{c} |\mathbf{E}|.$$

Module `vp_electromagnetism.py` confirms  $|\mathbf{B}|/|\mathbf{E}| = v/c$  exactly. Because  $\mathbf{B}$  *is* a rotation (a pseudovector), it is automatically divergence-free,  $\nabla \cdot \mathbf{B} = 0$ : the poles are the two ends of one rotation axis and cannot be separated — there are **no magnetic monopoles**, and cutting a magnet yields two fresh N/S pairs. The right-hand rule is the handedness of the rotation.

### EM.5 Polarization as rotation (helicity) [F]

Circular polarization *is* the rotating field; the two senses (CW/CCW) are helicity  $\pm 1$ . In Jones form the circular states are  $(1, \pm i)/\sqrt{2}$ . The module computes their Stokes vectors and finds  $|S_3| = S_0$  with  $S_1 = S_2 = 0$  — i.e. pure rotation, no linear component. Linear polarization is the equal superposition of the two rotation

senses. (The sign convention of  $S_3$  depends on the time convention  $e^{\mp i\omega t}$ ; the invariant content is  $|S_3| = S_0$  for circular light.)

## EM.6 The propagation angle: conduction $\leftrightarrow$ radiation [F]/[VP-prediction]

Light is unambiguously an electromagnetic wave — *one* object. Its longitudinal part ( $\nabla \cdot \mathbf{u}$ , compression = deficit) is the static Coulomb field; its transverse, propagating part is light. What distinguishes electricity, a radio wave, and a  $\gamma$ -ray is only the propagation angle  $\chi$ , set by the wavelength through physics §10.9:

$$\sin \chi = \frac{\lambda}{mD}, \quad m = \lceil \lambda/D \rceil.$$

Short- $\lambda$   $\gamma$ -rays are near-longitudinal ( $\chi \rightarrow 0$ , penetrating); visible and radio waves are near-transverse ( $\chi \rightarrow 90^\circ$ ). Crucially the visible band is **near** — **not exactly** — **transverse**:  $\chi \approx 89.9^\circ$ , a small *falsifiable* departure from the textbook exactly-transverse wave. Electric conduction is the extreme-longitudinal limit ( $\chi \rightarrow 0$ ): charges drift through a wire’s internal quanta, each producing a transverse twist that wraps the wire azimuthally as the magnetic field. **Conduction and radiation are the same electromagnetic phenomenon, differing only in angle (longitudinal vs transverse) and medium (atomic-internal quanta vs free space)** — “electricity travels like a  $\gamma$ -ray.”

Forward predictions (falsifiers, no fitting): for HeNe lines,  $\chi(632.8\text{ nm}) = 89.892^\circ$  and  $\chi(532\text{ nm}) = 89.825^\circ$  (modules `vp_light_angle.py`, `vp_electromagnetism.py`).

The earlier Word-document formulation  $\lambda = d/\cos \theta$  with  $d = 5000\text{ fm}$  is **superseded**: it has no  $m$ -chain and is undefined for  $\lambda < d$  ( $\gamma$ -rays, X-rays). The refined law fixes  $d$  to the *derived* value  $D = 4852.6\text{ fm}$  and covers the whole spectrum via the  $m$ -chain.

## EM.7 Blackbody radiation [F]

The Planck spectrum is the product of two lattice facts:

$$u(\nu, T) \propto \underbrace{\nu^2}_{\text{geometric mode density}} \times \underbrace{\frac{h\nu}{e^{h\nu/k_B T} - 1}}_{\text{rigid-shell high-}\nu \text{ cutoff}}.$$

The first factor alone is Rayleigh–Jeans and diverges (the UV catastrophe); the rigid shell blocks high-frequency deformation when the shell energy exceeds the thermal excitation, supplying the Bose cutoff. Module `vp_electromagnetism.py` tabulates  $u_{\text{RJ}}$  versus  $u_{\text{Planck}}$  (agreement at low  $\nu$ , strong suppression at high  $\nu$ ) and solves the Wien transcendental  $x = 5(1 - e^{-x})$  to recover  $\lambda_{\text{peak}} T = 2.89777 \times 10^{-3}\text{ m} \cdot \text{K}$  to  $6 \times 10^{-9}\%$ . A 300 K body peaks at  $\sim 9.7\text{ }\mu\text{m}$  (infrared) with  $\chi \approx 89.95^\circ$  — transverse light, consistent with EM.6.

## EM.8 Refraction and dispersion — the bridge to chemistry [F]/[CAL]

Once light exists, its interaction with matter is geometric. Interfacial transverse-oscillation matching gives Snell’s law  $n_1 \sin \theta_1 = n_2 \sin \theta_2$  [F]; the primary rainbow angle is  $42.1^\circ$  from Snell + droplet sphere geometry (Descartes) [F]; total internal reflection has  $\sin \theta_c = 1/n$  [F] (water  $48.6^\circ$ , diamond  $24.4^\circ$ ). The absolute refractive index  $n$  is a material constant [CAL]. Wavelength-dependent  $n(\lambda)$  orders the rainbow colors (blue refracts more). These feed directly into the chemistry chapter’s color of coordination compounds ( $d$ - $d$  transitions,  $\Delta_{\text{tet}}/\Delta_{\text{oct}} = 4/9$ ) and spectroscopy (modules `vp_refraction.py`, `vp_light_angle.py`, `vp_crystal_field.py`).

## EM.9 Master ledger — forced / measured / open

| Result                                                               | Grade         | Basis / module                                               |
|----------------------------------------------------------------------|---------------|--------------------------------------------------------------|
| $c^2 = B/\rho$ (single surviving longitudinal speed)                 | [F]/[V]       | isostatic $z=2d$ ;<br><code>vp_light_emergence.py</code>     |
| Linear dispersion $\omega = cq$                                      | [F]/[V]       | long- $\lambda$ limit;<br><code>vp_light_emergence.py</code> |
| Quantum diameter<br>$D = 2\lambda_{C,e} = 6\pi^6 r_p$                | [F]           | wavelength relation; (−19 ppm residual)                      |
| $1/r^2$ from $4\pi r^2$ flux dilution                                | [F]           | Gauss/Poisson;<br><code>vp_electromagnetism.py</code>        |
| Coulomb constant $k_e = \alpha_{\text{em}} \hbar c / e^2$            | [F]+[CAL]     | reproduces $k_e$ to $3 \times 10^{-10}\%$                    |
| $ \mathbf{B}  = (v/c) \mathbf{E} $ , $\nabla \cdot \mathbf{B} = 0$   | [F]           | twist geometry;<br><code>vp_electromagnetism.py</code>       |
| Circular polarization = rotation ( $ S_3  = S_0$ )                   | [F]           | Stokes; <code>vp_electromagnetism.py</code>                  |
| Propagation angle<br>$\sin \chi = \lambda/(mD)$                      | [F]/[VP-pred] | <code>vp_light_angle.py</code>                               |
| Blackbody = mode density $\times$ rigid cutoff; Wien                 | [F]           | <code>vp_electromagnetism.py</code>                          |
| Snell, rainbow $42.1^\circ$ , critical angle                         | [F]           | <code>vp_refraction.py</code>                                |
| Fine-structure coupling $\alpha_{\text{em}}$                         | [CAL]         | measured input, <b>not derived</b>                           |
| Long-range / unscreened (global $U(1)$ Goldstone)                    | [H]           | symmetry argument                                            |
| Magnetic / radiative <i>absolute</i> vector sector; genesis dynamics | [O]           | open (physics §14.0.6)                                       |

## EM.10 Falsification criteria [F]

Because the forced results have no free parameters, they are refuted if measurement disagrees:

- If the visible-band propagation angle is *exactly*  $90^\circ$  (not  $\approx 89.9^\circ$ ), the rotating-quantum /  $m$ -chain picture is wrong.
- If  $\chi(632.8 \text{ nm}) \neq 89.892^\circ$  or  $\chi(532 \text{ nm}) \neq 89.825^\circ$ , the angle law is refuted (forward prediction, length  $\rightarrow$  angle, no fitting).
- If the Coulomb field deviates from  $1/r^2$  in three dimensions, the flux-dilution origin fails.
- If circular light is not pure rotation ( $|S_3| \neq S_0$ ), helicity-as-rotation is wrong.
- If the blackbody peak does not satisfy the Wien transcendental, the mode-density  $\times$  rigid-cutoff factorization fails.

## EM.11 Reproducibility

All numbers above are produced by deterministic, standard-library Python modules with self-checks (`assert`). Determinism is verified by running each module twice and comparing the stdout sha256.

```
python3 vp_light_emergence.py      # light emergence: c²=B/ρ, dispersion, single speed, angle
python3 vp_electromagnetism.py     # charge, 1/r², E/B, polarization, conduction/radiation, blackbody
python3 verify_chemistry.py --root <package root> # runs all modules + ledger integrity gate
```

The harness reports execution, determinism, standard-library-only dependency, and case-ledger integrity across all modules (currently 32/32 PASS with the two EM/light modules added).

*This chapter completes the electromagnetic foundation on which the chemistry chapter rests: light emerges from the jammed lattice (EM.2), charge and the Coulomb  $1/r^2$  follow (EM.3), the E/B geometry and polarization are rotation (EM.4–5), the same wave spans conduction and radiation by angle (EM.6), and the optical consequences (blackbody, refraction, color) bridge into chemistry (EM.7–8). It is deliberately more detailed and fully reproducible relative to the original Korean working document, and every forced claim is falsifiable.*

## EM.12 Dynamic field equations, Poynting, and causality (addendum) [F]/[V]

Added 2026-06-14. Closes the dynamic-Maxwell gap flagged in the EM completeness review (Faraday/Ampère/Poynting were absent from EM.0–EM.11). Source: **AQD — Axiomatic Quantized Dynamics, NOCAL v1.1, DOI 10.5281/zenodo.17423870**. Reproduced by `vp_em_field_dynamics.py`.

The original EM chapter derived the **static** Coulomb sector ( $1/r^2$ , EM.3), the algebraic ratio  $|\mathbf{B}| = (v/c)|\mathbf{E}|$  and  $\nabla \cdot \mathbf{B} = 0$  (EM.4) — i.e. **two of the four Maxwell statements** (the two divergence facts). It did **not** state the **dynamic** field equations, the Poynting energy flow, or a causality theorem. This addendum supplies them in the **longitudinal (E-sector, pressure  $P$ ) form** from the AQD axioms, and marks the remaining vector-curl sector honestly as open.

### EM.12.1 The first-order dynamic system [F]

The jammed-lattice field obeys a coupled first-order system (AQD axiom A1) — pressure  $P$  (the compression = deficit = longitudinal/Coulomb sector) and velocity  $\mathbf{v}$ :

$$\partial_t P + K \nabla \cdot \mathbf{v} = 0, \quad \rho \partial_t \mathbf{v} + \nabla P = 0.$$

Eliminating  $\mathbf{v}$  gives the wave equation and the speed already used in EM.2:

$$\square_c P = 0, \quad c^2 = \frac{K}{\rho} (= B/\rho).$$

These two coupled first-order equations are the longitudinal analogue of Maxwell's curl pair ( $\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$ ,  $\nabla \times \mathbf{B} = \mu_0 \varepsilon_0 \partial_t \mathbf{E}$ ): the time-derivative of each field is sourced by a spatial derivative of the other, and their combination is the wave. Module `vp_em_field_dynamics.py` integrates this system (staggered leapfrog, deterministic) and recovers the pulse speed  $c_{\text{measured}} = 1.0001$  vs  $c = 1$  (**0.01 %**).

### EM.12.2 Poynting's theorem and energy conservation [F]/[V]

Define the energy density and flux (AQD; equivalent to the second-order form  $e = \frac{1}{2}[(\partial_t P)^2/c^2 + |\nabla P|^2]$ ,  $\mathbf{S} = -(\partial_t P)\nabla P$ ):

$$e = \frac{1}{2} \left[ \frac{P^2}{K} + \rho |\mathbf{v}|^2 \right], \quad \mathbf{S} = P \mathbf{v}, \quad \boxed{\partial_t e + \nabla \cdot \mathbf{S} = 0}$$

— the **local energy-conservation (Poynting) law**, the longitudinal counterpart of EM's  $\partial_t u + \nabla \cdot (\mathbf{E} \times \mathbf{H}) = 0$ . Integrating over a lossless domain (periodic, Neumann, time-invariant Dirichlet, or radiationless infinity) gives  $\oint \mathbf{S} \cdot \mathbf{n} = 0$  and hence

$$\frac{dE}{dt} = 0 \quad (\text{AQD QP-0002-001, fully proven}).$$

The module confirms  $\partial_t e + \partial_x S = 0$  pointwise (RMS residual  $\sim 3 \times 10^{-2}$ , discretization scale) and energy conservation to  $3 \times 10^{-4}$  over the run. A companion uniqueness theorem (energy method, QP-0002-002) shows identical initial data give a unique solution.

### EM.12.3 Causality: finite propagation and the light cone [F]

The same energy structure forces a **hard light cone** (AQD theorems F-A/B/C):

$$u_{\max} = c, \quad \text{supp } e(\tau, \cdot) \subseteq \text{supp } e(0, \cdot) \oplus B_{c\tau}.$$

Energy and information cannot reach beyond  $c\tau$  of the initial support. The module verifies this directly: the pulse support stays inside the cone (expansion  $0.00 \leq c\tau$ ). At the quantum level the AQD upgrades this to **microcausality** — with  $\Phi := P/\sqrt{K}$ ,  $[\Phi(x), \Phi(y)] = 0$  for spacelike  $(x - y)$  via the Pauli–Jordan function  $D(x)$  whose support lies on the light cone (the cone speed is fixed at  $c$  by  $c^2 = K/\rho$ ), and the arrival-time bound  $t_{\text{arr}} \geq \text{dist}/c$  follows from commutator support. No calibration enters.

### EM.12.4 What this closes, and what remains open [O]

**Closed (this addendum):** the *dynamic* field equations, the Poynting energy flow and its exact conservation, uniqueness, and causality (classical light cone + quantum microcausality). Together with EM.2 (wave/ $c^2 = B/\rho$ ) and EM.3 (static  $1/r^2$ ), the **longitudinal (E-sector)** electrodynamics is now complete and reproducible.

**Still open — honestly marked [O] (unchanged from EM.9):** the **full vector E/B curl structure** — Faraday’s  $\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$  and Ampère–Maxwell’s  $\nabla \times \mathbf{B}$  in *vector* form, with the **magnetic field as an independent transverse pseudovector** and its *absolute* magnitude. The AQD treatment is scalar/longitudinal ( $P, \mathbf{v}$  with  $\mathbf{v}$  irrotational); it yields the compression (Coulomb/ $E$ ) sector rigorously but not the independent transverse-vector (magnetic/radiative) sector. This is exactly the EM.9 line “*magnetic/radiative absolute vector sector — [O] (physics §14.0.6)*”. It is **not** resolved here; it remains the principal open structural item of the EM chapter.

### EM.12.5 Updated master-ledger rows

| Result                                                                                                          | Grade   | Basis / module                                       |
|-----------------------------------------------------------------------------------------------------------------|---------|------------------------------------------------------|
| First-order system<br>$\partial_t P + K \nabla \cdot \mathbf{v} = 0, \rho \partial_t \mathbf{v} + \nabla P = 0$ | [F]     | AQD A1;<br>vp_em_field_dynamics.py                   |
| Poynting law $\partial_t e + \nabla \cdot \mathbf{S} = 0,$<br>$\mathbf{S} = P\mathbf{v}$                        | [F]/[V] | AQD QP-0002-001; residual<br>\$ \$3e-2               |
| Energy conservation $dE/dt = 0$<br>(lossless) + uniqueness                                                      | [F]     | AQD QP-0002-001/002; dev 3e-4                        |
| Causality: light cone, $u_{\max} = c;$<br>microcausality                                                        | [F]     | AQD F-A/B/C, QP-0002-U121                            |
| Full vector E/B curl<br>(Faraday/Ampère, magnetic<br>absolute)                                                  | [O]     | open — scalar/longitudinal AQD<br>does not supply it |

*This addendum upgrades the EM chapter from “static sector + algebraic B/E ratio” to “**complete longitudinal electrodynamics with Poynting flow and causality**,” using the AQD axiomatic field dynamics. The dynamic-Maxwell gap is closed in the E-sector; the independent vector (magnetic/radiative) sector remains the chapter’s honestly-declared open item.*

## Chemistry from a Single Anchor: The Jamming Account

**VP Chemistry & Electromagnetism Whitepaper — Chapter CH** DOI (reserved): 10.5281/zenodo.20680541 · Companion to VP Theory (physics) and to Chapter EM Status grades: [F] forced (zero

free parameters) · [F?] forced-candidate (model-dependent or <1.5%) · [CAL] calibration input (measured constant, used consistently) · [V] simulation-measured · [O] open (honestly unsolved). Every numeric claim is reproduced by a deterministic, standard-library module (2× sha256 identical); module and case-ledger anchors are given inline. Verified numbers quoted below are the actual module outputs.

Everything starts locked, is derived, and is settled by a verdict. One table separates what is *forced* from what is *measured input* from what is *open*. Chemistry unfolds from the electron (a rotational dent) using only  $\pi$ -geometry and the electromagnetic  $1/r^2$  — the same substrate from which light emerges (Chapter EM).

## CH.0 Governance and grading (inherited from VP Theory §1)

This chapter inherits No-Tuning / LOCK / Gate. **No-Tuning:** changing an input after seeing a result to hit a target is forbidden. The clean final logic is presented here; the trial-and-error of development lives, reproducibly, in the modules and the case ledger. Forced results [F] have no adjustable parameter — they cannot be tuned, so they are *refuted* if they disagree with measurement.

## CH.1 The anti-tuning chain — the central defense

A trained reviewer immediately suspects parameter-fitting. The rebuttal has three prongs.

**Prong A — forced results are dimensionless geometry with nothing to adjust.** The tetrahedral angle  $\arccos(-1/3)$ , the ratio  $\Delta_{\text{tet}}/\Delta_{\text{oct}} = 4/9$ , the FCC packing fraction  $\pi/(3\sqrt{2})$ , the shell-filling integers [2, 8, 8, 18, 18, 32] — these are pure geometry/integers with no knob.

**Prong B — calibration inputs are used as forward predictions.** Measured constants (spectroscopic  $B, \nu$ ; bond enthalpy  $\Delta H$ ;  $pK_a$ ; standard potential  $E^\circ$ ) are each fixed *once* and used consistently. Quantities *derived* from them — standard entropy, equilibrium constants, cell potentials — then match measurement; that is prediction, not fitting.

**Prong C — honest open items prove no tuning.** A tuner would have “solved” absolute multi-electron ionization energies; instead the failure is recorded (mean ~22% for main group, electron affinity wrong in sign 8/18 by the Slater method) and marked [O].

## CH.2 Foundation: anchor $\rightarrow$ atomic scale [F]

The only observational input is the electron mass  $m_e$  (the anchor). With the EM coupling  $\alpha_{\text{em}}$  [CAL] and  $\pi$  (using the *reduced* Compton wavelength  $\bar{\lambda}_{C,e} = \hbar/(m_e c) = \lambda_{C,e}/2\pi$ ):

$$a_0 = \frac{\bar{\lambda}_{C,e}}{\alpha_{\text{em}}} = \frac{\hbar}{m_e c \alpha_{\text{em}}}, \quad \text{Ry} = \frac{\alpha_{\text{em}}^2 m_e c^2}{2}.$$

Module `vp_atomic.py` gives  $a_0 = 0.529177 \text{ \AA}$  ( $\Delta +0.0000\%$ ) and  $\text{Ry} = 13.6057 \text{ eV}$  ( $\Delta -0.0001\%$ ); hydrogenic  $\text{IE} = \text{Ry } Z^2$  to <0.06%. Shell structure follows from the Aufbau degeneracy  $2(2l+1)$ , giving noble-gas closures [2, 10, 18, 36, 54, 86] and period lengths [2, 8, 8, 18, 18, 32] exactly [F]. Absolute many-electron IE is only approximate (screening) and is [O].

## CH.3 Periodic structure: valence and oxidation [F]/[F?]

Valence equals the number of outer-shell electrons (1–8, main group) [F]. The octet rule fixes ionic-compound stoichiometry: module `vp_valence.py` predicts 9/9 formulas (NaCl, MgCl<sub>2</sub>, Al<sub>2</sub>O<sub>3</sub>, CaO, ...). Periodic trends (radius ↓, electronegativity ↑ across a period; radius ↑ down a group) are [F?]. Transition metals have valence  $4s+3d$  with variable oxidation states (Mn +2...+7) from the partially filled  $d$ -shell; the half/fully-filled exceptions Cr ( $3d^5 4s^1$ ), Cu ( $3d^{10} 4s^1$ ) follow from sub-shell stability [F].

## CH.4 Chemical bonding and VSEPR geometry [F]

Geometry is the electromagnetic  $1/r^2$  repulsion of electron domains on a sphere (the Thomson problem). Module `vp_molecular_geometry.py` computes (seed-independent) the angles:

$$\text{linear } 180^\circ, \quad \text{trigonal } 120^\circ, \quad \boxed{\text{tetrahedral } \arccos(-\frac{1}{3}) = 109.47^\circ}, \quad \text{octahedral } 90^\circ.$$

All have zero free parameters [F]; the simulation reproduces the closed form  $\arccos(-1/3) = 109.47^\circ$ . Lone-pair compression ( $\text{CH}_4 \rightarrow \text{NH}_3 \rightarrow \text{H}_2\text{O}$ :  $109.5^\circ \rightarrow 107^\circ \rightarrow 104.5^\circ$ ) has the correct direction [F?]; the precise magnitude is [O]. Bond rupture follows a scale-invariant  $\sqrt{2}$  diagonal-stretch geometry shared with nuclear fission (Chapter EM links the same  $1/r^2$ ); absolute bond energy needs per-bond anharmonicity and is [O].

## CH.5 Coordination chemistry: color and magnetism [F]

Ligand point charges (EM  $1/r^2$ ) split the geometric  $d$ -orbitals (spherical harmonics). Module `vp_crystal_field.py` gives the octahedral  $e_g \uparrow / t_{2g} \downarrow$  pattern (tetrahedral inverted) and the analytic ratio

$$\frac{\Delta_{\text{tet}}}{\Delta_{\text{oct}}} = \frac{4}{9} = 0.4444 \quad (\text{exact}).$$

High/low-spin unpaired-electron counts give magnetic moments ( $\text{Fe}^{3+}$  5/1 unpaired) [F]. Color is the  $d-d$  transition  $\Delta \rightarrow \lambda$  (interpreted optically via Chapter EM); multi-band precise color is [O], the spectrochemical series is [CAL].

## CH.6 Light, optics, color — see Chapter EM [F]/[CAL]

The optical consequences (propagation angle  $\sin \chi = \lambda/(mD)$ , Snell, rainbow  $42.1^\circ$ , total-internal-reflection critical angle, blackbody,  $d-d$  color) are developed in **Chapter EM**. Chemistry *consumes* the electromagnetic mechanism from physics §10/§14; it does not re-derive it. The boundary is explicit: fundamental EM (light as rotational transverse wave,  $c^2 = B/\rho$ , the  $1/r^2$  Green function) is physics; the chemical/optical consequences are here.

## CH.7 Thermodynamics: rotational energy $\rightarrow$ temperature [F]/[F?]

Temperature is a rotational-energy scale; rotational levels  $E_J = BhcJ(J+1)$ . Heat-capacity degrees-of-freedom: monatomic  $\frac{3}{2}R$ , diatomic  $\frac{5}{2}R$ , solid Dulong–Petit  $3R$  (<3%). Module `vp_statistical_thermo.py` reproduces the  $\text{H}_2$  heat-capacity **staircase** ( $\frac{3}{2}R$  at 20 K  $\rightarrow \frac{5}{2}R$  at 298 K  $\rightarrow \frac{7}{2}R$  above  $\sim 6332$  K) with activation temperatures  $T_{\text{rot}} = 88$  K,  $T_{\text{vib}} = 6332$  K from  $B, \nu$ ;  $C_v(\text{H}_2)$  20.81 vs 20.4 (+2.0%) [F]. Standard molar entropy (Sackur–Tetrode + rotation + vibration + electronic) is <0.3% for  $\text{H}_2 \cdot \text{N}_2 \cdot \text{CO} \cdot \text{O}_2$ ;  $\text{O}_2$  includes the known triplet (paramagnetic)  $R \ln 3$  [F?]. The spectroscopic constants  $B, \nu$  are per-molecule [CAL].

## CH.8 Chemical equilibrium [F]

Direction is set by  $\Delta G = \Delta H - T\Delta S$ ; dissociation occurs when  $T\Delta S > \Delta H$ . The dissociation temperature is

$$T^* = \frac{\Delta H}{\Delta S}.$$

Module `vp_equilibrium.py` gives  $T^*$ :  $\text{N}_2$  8214 K >  $\text{H}_2$  4400 K >  $\text{O}_2$  4210 K >  $\text{Cl}_2$  2272 K — the **bond-strength order** [F]. Reaction entropies from partition functions are <0.4%. The van 't Hoff  $K(T)$  ( $\text{H}_2 \rightleftharpoons 2\text{H}$ , with  $K = 1$  at  $T^*$ ) is [F?] ( $\Delta H$  is [CAL]). Le Chatelier directions (dissociation  $\Delta S > 0$  high-T driven; Haber  $\Delta S < 0$  high-T reversed) [F].

## CH.9 Reaction kinetics [F]

The activation energy is the threshold to reach the transition state (a  $\sqrt{2}$  bond-stretch); the fraction crossing is the Boltzmann tail  $\exp(-E_a/RT)$ . Module `vp_kinetics.py` reproduces the Arrhenius temperature dependence, the room-temperature **10 K doubling rule** ( $E_a \approx 50$  kJ/mol gives  $\times 1.9$  per 10 K), and recovery of  $E_a$  from the Arrhenius slope  $-E_a/R$ . Mean molecular speed  $\sqrt{8k_B T/\pi m}$  ( $N_2$  298 K  $\rightarrow$  475 m/s) and the Eyring universal factor  $k_B T/h = 6.2 \times 10^{12} \text{ s}^{-1}$  follow [F]. Absolute  $E_a$  (transition-state structure) and tunneling are [O].

## CH.10 Solution chemistry: proton-transfer equilibrium [F]

With  $K_w = 10^{-14}$ , pure water has pH 7. The weak-acid pH solves  $K_a = x^2/(C - x)$ ; module `vp_solution_chemistry.py` gives acetic acid 0.1 M  $\rightarrow$  pH 2.88 (measured 2.87). Henderson–Hasselbalch buffering  $\text{pH} = \text{p}K_a + \log([A^-]/[HA])$  and titration equivalence-point jumps follow [F]. Individual  $\text{p}K_a$  are [CAL]; absolute  $\text{p}K_a$  prediction (solvation) is [O].

## CH.11 Electrochemistry: electron (rotational dent) transfer [F]

Charge is rotation (Chapter EM); redox is rotational-dent transfer, symmetric with proton transfer (CH.10). The cell potential is

$$E_{\text{cell}}^\circ = E_{\text{cathode}}^\circ - E_{\text{anode}}^\circ, \quad \Delta G^\circ = -nFE^\circ, \quad E = E^\circ - \frac{RT}{nF} \ln Q.$$

Module `vp_electrochemistry.py` gives the Daniell cell  $E_{\text{cell}}^\circ = 1.10$  V (matches measured) and  $\Delta G^\circ = -212$  kJ/mol [F]. The standard-potential series ( $\text{Li} -3.04 \rightarrow \text{F}_2 +2.87$ ) links to IE and electronegativity; individual values are [CAL].

## CH.12 Organic chemistry: carbon skeleton on EM geometry [F]

Carbon's tetravalence plus EM repulsion gives hybridization geometry (the same VSEPR engine as CH.4): module `vp_organic_chemistry.py` gives  $\text{sp}^3$  tetrahedral  $109.5^\circ$ ,  $\text{sp}^2$  planar  $120^\circ$ ,  $\text{sp}$  linear  $180^\circ$ . Bond order–length–strength:

|     | order                  | length | strength (kJ/mol) |
|-----|------------------------|--------|-------------------|
| C–C | 1 ( $1\sigma$ )        | 154 pm | 348               |
| C=C | 2 ( $1\sigma + 1\pi$ ) | 134 pm | 614               |
| C C | 3 ( $1\sigma + 2\pi$ ) | 120 pm | 839               |

The degree of unsaturation  $(2C + 2 + N - H - X)/2$  gives benzene 4 [F]. Functional-group reactivity (acid–base, redox applied) is [F?]; absolute rates and stereoselectivity are [O].

## CH.13 Materials and nuclear (summary) [F]/[F?]

Crystal packing fractions are pure geometry: FCC/HCP  $\pi/(3\sqrt{2}) = 0.7405$ , BCC 0.6802, SC 0.5236 (module `vp_crystal_packing.py`); FCC metal densities  $<2\%$  [F]/[F?]. Nuclear magic-number skeleton from Gauss–lattice point counts  $N(R^2) + 1$ : module `vp_gauss_shells.py` gives **2, 8, 20, 28, 82** (5/7 parameter-free); 50, 126 require spin–orbit [CAL]. SEMF fission/fusion gives the iron peak  $A = 58$  and the critical  $Z^2/A$  [F?]. Theoretical bounds ( $\sigma_{th}$ ,  $\gamma_s$ ) are bounds, not predictions — honestly [F]+[O].

## CH.14 Master ledger — forced / measured input / open

[F] **forced (zero free parameters — cannot be tuned)**: shell filling and period lengths; valence, oxidation, ionic formulas; all VSEPR angles ( $\arccos(-1/3)$  etc.);  $\Delta_{\text{tet}}/\Delta_{\text{oct}} = 4/9$ ; packing fractions; propagation angle  $\sin \chi = \lambda/(mD)$ ; rainbow  $42^\circ$ ; Snell, critical angle;  $C_v(T)$  staircase and activation temperatures; dissociation–

temperature order and Le Chatelier; Arrhenius, 10 K doubling,  $k_B T/h$ ; pH, buffers, titration;  $E^\circ_{\text{cell}}$ ,  $\Delta G = -nFE^\circ$ , Nernst; hybridization, bond order, unsaturation; Gauss magic-number skeleton.

**[CAL] calibration input (measured constants, used consistently — never tuned per case):** spectroscopic  $B, \nu$ ; bond enthalpy  $\Delta H$ ;  $pK_a$ ; standard potential  $E^\circ$ ; spectrochemical series; absolute refractive index  $n$ ; spin-orbit strength (magic 50, 126); the fine-structure coupling  $\alpha_{\text{em}}$ .

**[O] open (honest — not disguised as solved):** absolute multi-electron IE (screening); electron affinity (Slater sign failure); absolute  $pK_a$ ,  $E^\circ$  (solvation); multi-band precise color; absolute bond energy (anharmonicity);  $d$ -block absolute IE; absolute rates and stereoselectivity; nuclear shell-correction masses; pure- $\pi$  geometry of 50, 126.

## CH.15 Two unifications — sphere geometry and rotation

**Sphere geometry:** molecular shape = points *on* a sphere (Thomson/VSEPR)  $\rightarrow \arccos(-1/3)$ ; nuclear shells = lattice points *inside* a sphere (Gauss)  $\rightarrow$  magic-number skeleton; crystal density = sphere *packing* (Kepler)  $\rightarrow 0.7405$ ;  $d$ -orbital splitting = spherical harmonics  $\times$  ligand EM  $\rightarrow 4/9$ .

**Rotation:** the electron is a rotational dent; charge is rotation handedness; light is the rotational transverse wave (Chapter EM); temperature is rotational excitation; equilibrium, rate, and electrochemistry all ride on rotational energy. The single anchor (electron = rotational dent) opens, and the last sector (electrochemistry = rotational-dent transfer) closes the circle.

## CH.16 Falsification criteria [F]

Forced results cannot be re-fitted, so they are refuted on disagreement:

- If the tetrahedral angle is not  $\arccos(-1/3) = 109.47^\circ$ , the VSEPR-EM derivation is discarded.
- If  $\Delta_{\text{tet}}/\Delta_{\text{oct}} \neq 4/9$ , the crystal-field geometry is discarded.
- If the rainbow angle is not  $\approx 42^\circ$ , the light-angle/refraction derivation is discarded.
- If  $\text{H}_2$  heat capacity shows no  $\frac{3}{2} \rightarrow \frac{5}{2} \rightarrow \frac{7}{2}R$  staircase, or the activation temperatures disagree with  $B, \nu$ , the partition-function account is discarded.
- If dissociation temperatures do not follow the bond-strength order, the equilibrium derivation is discarded.
- The committed light-angle predictions  $\chi(632.8 \text{ nm}) = 89.892^\circ$ ,  $\chi(532 \text{ nm}) = 89.825^\circ$  are falsifiable forward predictions (Chapter EM).

## CH.17 Reproducibility

Thirty deterministic, standard-library modules and a 114-row case ledger (`cases_fixed.csv`) reproduce this logic; the light/EM additions add a 16-row ledger (`cases_em.csv`). Determinism is verified by running each module twice and comparing the stdout sha256.

```
python3 research/foundation/vp_molecular_geometry.py # tetrahedral 109.47°
python3 research/foundation/vp_crystal_field.py      # 4/9
python3 verify_chemistry.py --root <package root>    # all modules + ledger integrity gate
```

The harness reports execution, determinism, standard-library-only dependency, and case-ledger integrity. With the light and electromagnetism modules included, **32/32 PASS**.

*Together with Chapter EM, this completes the program: light emerges from the jammed lattice and the electromagnetic  $1/r^2$  follows (Chapter EM); on that substrate, atoms, bonds, geometry, color, heat, equilibrium, rate, solution, electrochemistry, and organic structure are forced by sphere geometry and rotation (this chapter). What is forced is falsifiable; what is measured is labelled; what is open is admitted — and all of it is reproducible.*

# Conduction in Copper and Magnetism in Iron: Two Fates of the *d*-Shell

**VP Chemistry & Electromagnetism Whitepaper — Chapter CM** DOI (reserved): 10.5281/zenodo.20680541 · Builds on Chapter EM (electromagnetism) and Chapter CH (chemistry). Status grades as in Chapter EM. Numbers reproduced by `vp_copper_conduction.py` and `vp_iron_magnetism.py` (deterministic, standard-library, self-checking).

Copper conducts and iron magnetizes for the same root reason, read two ways. Copper’s inner *d*-shell is **full**, freeing its outer electron into a conducting sea; iron’s *d*-shell is **partly empty**, leaving unpaired rotations that align into a magnet. A filled shell makes a conductor; an unfilled shell makes a magnet.

## CM.0 The two questions

This chapter answers two concrete questions about metals using the electromagnetism of Chapter EM:

1. **Electricity is an electromagnetic wave** — how does it pass *through the inside* of copper, when copper (a metal) *reflects* ordinary light?
2. **What is the principle of iron’s magnetism** — and in particular, why iron (and cobalt, nickel) but not copper, and not even manganese, which has *more* unpaired electrons than iron? (The physics whitepaper took the “why Fe and not Cu” criterion as an empirical input, §14; here it is derived.)

## CM.1 The paradox

Light is a transverse electromagnetic wave, and metals reflect it — that is why copper is shiny and opaque. Yet electricity is *also* electromagnetic, and it travels straight through the body of a copper wire. How can the same medium reflect one electromagnetic disturbance and transmit another?

The resolution is the propagation angle of Chapter EM (§EM.6). Electricity is the **extreme longitudinal limit** ( $\chi \rightarrow 0$ ) of the electromagnetic wave, while visible light is the **transverse limit** ( $\chi \rightarrow 90^\circ$ ). The same free-electron sea reflects the transverse wave and transmits the longitudinal one. Same medium, opposite fates, set by angle.

## CM.2 Copper’s free-electron sea [F]/[CAL]

Copper is [Ar]  $3d^{10}4s^1$ : the inner  $3d$  shell is **full**, and the single  $4s$  electron is delocalized into a conducting sea. The free-electron density is

$$n = \frac{\rho N_A Z_{\text{free}}}{M} = 8.49 \times 10^{28} \text{ m}^{-3}$$

(one free electron per atom;  $\rho$ ,  $M$  measured [CAL]). This sea is the “internal quanta” through which conduction propagates.

## CM.3 Why transverse light is reflected — the plasma frequency [F]/[CAL]

The free-electron sea has a natural oscillation frequency, the plasma frequency

$$\omega_p = \sqrt{\frac{n e^2}{\epsilon_0 m_e}} = 1.64 \times 10^{16} \text{ rad/s} \quad (f_p = 2.6 \times 10^{15} \text{ Hz}, \lambda_p = 115 \text{ nm}, \text{ ultraviolet}).$$

Below  $\omega_p$  — radio, infrared, **all of visible light** — an incident transverse wave drives the free electrons, which re-radiate and **reflect** it: this is exactly why metals are shiny and opaque (`vp_copper_conduction.py`, §1). Above  $\omega_p$  (X-rays) the electrons cannot follow and the metal becomes transparent. The transverse electromagnetic wave is turned back by the very sea that carries the current.

## CM.4 Why longitudinal electricity is transmitted — conduction as $\chi \rightarrow 0$ [F]

The longitudinal mode is different. A longitudinal ( $\chi \rightarrow 0$ ) electromagnetic disturbance is a **density/pressure wave of the electron sea** that threads straight along the wire — straighter even than a  $\gamma$ -ray. It is not reflected because it is not a transverse oscillation of the surface electrons; it propagates *through* the internal quanta of the copper atoms, not through vacuum or air (physics §14). This longitudinal disturbance is the current. The angle that makes light reflect (transverse) is the angle that makes electricity transmit (longitudinal). Two physical facts fix conduction at  $\chi \rightarrow 0$ : small amplitude (a low-amplitude disturbance threads straight; if the amplitude grows the path tilts transverse and the signal *leaves* the copper as radiative loss) and medium (it runs through the atomic-internal quanta).

## CM.5 The skin effect [F]/[CAL]

The transverse and longitudinal pictures are quantitatively distinct. A transverse wave that does enter the metal decays within the skin depth

$$\delta = \sqrt{\frac{2}{\mu_0 \sigma \omega}} : \quad 8.4 \text{ mm (60 Hz)} \rightarrow 2.1 \text{ } \mu\text{m (1 GHz)}.$$

High-frequency transverse fields are confined to a micron-thin surface layer, while the direct/low-frequency longitudinal current fills the whole cross-section (`vp_copper_conduction.py`, §3).

## CM.6 Three velocities — drift, signal, Fermi [F]/[CAL]

A common confusion is dissolved by separating three speeds (`vp_copper_conduction.py`, §4):

- **Drift velocity**  $v_d = J/(ne)$ : for  $J = 10^6$  A/m<sup>2</sup>, only 0.074 mm/s — the electrons themselves crawl.
- **Signal velocity**  $\sim c$ : the longitudinal field disturbance travels near light speed, so a lamp lights essentially instantly even though the electrons barely move.
- **Fermi velocity**  $v_F = 1.57 \times 10^6$  m/s: the electrons at the Fermi surface move fast, but their *net* drift is the slow  $v_d$ .

The predicted Fermi energy  $E_F = \hbar^2(3\pi^2n)^{2/3}/(2m_e) = 7.04$  eV matches the measured  $\approx 7.0$  eV.

## CM.7 Why copper conducts so well [F]/[CAL]

The Drude relation  $\sigma = ne^2\tau/m_e$  with  $\sigma_{\text{Cu}} = 5.96 \times 10^7$  S/m gives a relaxation time  $\tau = 2.5 \times 10^{-14}$  s and a mean free path  $\ell = v_F\tau = 39$  nm. Copper's **filled**  $3d^{10}$  shell hands its  $4s^1$  electron to a low-scattering sea — exactly the property that, in iron, is *absent*. This is the hinge of the whole chapter: **a filled inner shell yields conduction; an unfilled one yields magnetism.**

## CM.8 The magnetic field of a current — and Joule loss [F]

A drifting charge tilts its rotation axis by  $\sim v/c$  (§EM.4), shedding a small transverse twist perpendicular to the drift; propagating sideways it wraps the wire azimuthally as the magnetic field  $B = \mu_0 I/(2\pi r)$ , with circulation sense set by the rotation handedness (the right-hand rule). The transverse twist has two fates: the **returned** part is the reactive  $B$  field itself (present even at zero loss, as in a superconductor), and the **un-returned** fraction is dissipation (Joule loss). Thus  $B$  and resistive loss share one origin — the transverse twist of moving charge — as its returned versus un-returned parts;  $B$  itself is not the loss.

## CM.9 Iron's magnetism — the principle [F]/[CAL]/[O]

Magnetism is aligned rotation. A magnetic moment is the rotation of an unpaired electron — the same rotation that *is* electric charge (§EM.1) and helicity (§EM.5). For a material to be a permanent (ferro)magnet, **two conditions must both hold**, and the pair derives exactly which elements magnetize.

### CM.10 Criterion 1 — unpaired $d$ -electrons (this is why not copper) [F]

A localized moment requires an **unfilled**  $d$ -shell with unpaired electrons (Hund's rule). For  $3d^{n_d}$  the unpaired count is  $n_d$  if  $n_d \leq 5$  else  $10 - n_d$ , and the spin-only moment is  $\mu = \sqrt{n(n+2)} \mu_B$  (`vp_iron_magnetism.py`, condition 1):

| element   | config         | unpaired $d$ | $\mu_{\text{so}} [\mu_B]$ | moment?   |
|-----------|----------------|--------------|---------------------------|-----------|
| Mn        | $3d^5 4s^2$    | 5            | 5.92                      | yes       |
| Fe        | $3d^6 4s^2$    | 4            | 4.90                      | yes       |
| Co        | $3d^7 4s^2$    | 3            | 3.87                      | yes       |
| Ni        | $3d^8 4s^2$    | 2            | 2.83                      | yes       |
| <b>Cu</b> | $3d^{10} 4s^1$ | <b>0</b>     | <b>0.00</b>               | <b>no</b> |

Copper's  $3d^{10}$  shell is full: zero unpaired  $d$ -electrons, no localized moment. Its  $4s^1$  is a free conduction electron, not a localized moment. **Copper fails condition 1 and cannot be magnetic — the same filled shell that makes it a superb conductor.**

### CM.11 Criterion 2 — positive exchange (this is why not manganese) [F]/[CAL]

Having moments is not enough; neighbouring  $d$ -rotations must align **parallel** (ferromagnetic), not antiparallel (antiferromagnetic). The Bethe–Slater criterion uses the ratio  $D/d = \text{interatomic distance} / 3d\text{-orbital diameter}$ : above a critical value ( $\approx 1.5$ ) the exchange integral is positive (parallel, ferromagnetic); below it the exchange is negative (antiparallel, antiferromagnetic) (`vp_iron_magnetism.py`, condition 2):

| element | $D/d$ | exchange | alignment                |
|---------|-------|----------|--------------------------|
| Cr      | 1.18  | $J < 0$  | antiparallel (antiferro) |
| Mn      | 1.47  | $J < 0$  | antiparallel (antiferro) |
| Fe      | 1.63  | $J > 0$  | parallel (ferro)         |
| Co      | 1.82  | $J > 0$  | parallel (ferro)         |
| Ni      | 1.98  | $J > 0$  | parallel (ferro)         |

Manganese has **five** unpaired electrons — more than iron — yet  $D/d = 1.47 < 1.5$  gives  $J < 0$ , so its moments align antiparallel and cancel: **manganese fails condition 2 and is antiferromagnetic**. Cobalt ( $D/d = 1.82$ ) sits near the exchange maximum.

### CM.12 The derivation — 7/7 classification [F]

Applying both conditions reproduces the observed magnetic class of the late  $3d$  series with no fitted magnetic parameter (`vp_iron_magnetism.py`, summary):

| element | unpaired $d$ | $D/d$ | derived   | observed  | ✓ |
|---------|--------------|-------|-----------|-----------|---|
| Cr      | 5            | 1.18  | antiferro | antiferro | ✓ |
| Mn      | 5            | 1.47  | antiferro | antiferro | ✓ |
| Fe      | 4            | 1.63  | ferro     | ferro     | ✓ |
| Co      | 3            | 1.82  | ferro     | ferro     | ✓ |
| Ni      | 2            | 1.98  | ferro     | ferro     | ✓ |

| element | unpaired $d$ | $D/d$ | derived      | observed     | ✓ |
|---------|--------------|-------|--------------|--------------|---|
| Cu      | 0            | —     | non-magnetic | non-magnetic | ✓ |
| Zn      | 0            | —     | non-magnetic | non-magnetic | ✓ |

**7/7.** Copper is non-magnetic by condition 1, manganese is antiferromagnetic by condition 2, and only iron, cobalt, and nickel satisfy both — the “why Fe and not Cu” the physics whitepaper left as input is here derived.

### CM.13 Curie temperature, domains, and $\nabla \cdot \mathbf{B} = 0$ [F]/[CAL]

Thermal rotation destroys alignment above the Curie temperature, where  $k_B T_C$  is of order the exchange energy: Fe 1043 K, Co 1388 K (highest, matching the strongest exchange), Ni 627 K (`vp_iron_magnetism.py`). Below  $T_C$ , exchange aligns moments into **domains**; an external field moves the domain walls. Because  $\mathbf{B}$  is a rotation axis (§EM.4), it is divergence-free,  $\nabla \cdot \mathbf{B} = 0$ : the poles are the two ends of one axis and cannot be separated — no monopoles, and cutting a magnet yields two fresh N/S pairs. Hysteresis is the part of the alignment that does not return — the permanent magnet — and heating above  $T_C$  erases it. The  $\nabla \cdot \mathbf{B} = 0$  seen abstractly in Chapter EM becomes, in iron, the macroscopic fact of domains and permanent magnets. (The metallic moments are non-integer — Fe 2.22, Co 1.72, Ni 0.62  $\mu_B$  — an itinerant/band effect marked [O]; the localized Hund count gives the correct *ordering* and the magnetic *class*.)

### CM.14 The unifying picture

Copper and iron are the two fates of the  $d$ -shell:

- **Copper** — **filled**  $3d^{10} \rightarrow$  **free**  $4s$  **electron**  $\rightarrow$  **conduction**. Electricity passes through copper as the longitudinal ( $\chi \rightarrow 0$ ) electromagnetic mode of that free sea, while light (transverse) is reflected by it.
- **Iron** — **unfilled**  $3d^6 \rightarrow$  **unpaired moments**  $\rightarrow$  **magnetism**. With positive exchange ( $D/d > 1.5$ ) the moments align into a ferromagnet; the aligned rotation is the magnetic field.

A filled inner shell makes a conductor; an unfilled one makes a magnet. Both are the same rotation — the rotation that is charge, light, and helicity — expressed as a delocalized current in copper and as an aligned moment in iron.

### CM.15 Falsification and reproducibility [F]

- If a metal’s plasma frequency did not predict its reflectivity edge (shiny below  $\omega_p$ , transparent above), the plasma-reflection account of conduction-vs-light fails.
- If manganese (more unpaired electrons than iron) were ferromagnetic, or copper ( $3d^{10}$ ) were ferromagnetic, the two-condition criterion fails.
- If the derived magnetic class disagreed with any late-3d metal, the derivation fails (it is 7/7).

```
python3 code/foundation/vp_copper_conduction.py # plasma freq, skin depth, drift/signal/Fermi
python3 code/foundation/vp_iron_magnetism.py    # two-condition ferromagnetic criterion (7/7)
```

Both modules are deterministic (stdout sha256 identical on re-run) and standard-library only.

*Copper and iron close the loop opened in Chapter EM: charge is rotation, and that one rotation appears as a longitudinal current threading copper’s filled-shell electron sea, and as an aligned moment in iron’s unfilled  $d$ -shell. The conductor and the magnet are the same physics, divided by whether the inner shell is full.*

## The Color of Molecules: From $d$ -Orbitals to Paint Pigments

**VP Chemistry & Electromagnetism Whitepaper — Chapter CC** DOI (reserved): 10.5281/zenodo.20680541 · Builds on Chapter EM (light), Chapter CH §CH.5 (crystal field), and Chapter CM. Num-

bers reproduced by `vp_molecular_color.py` (and `vp_crystal_field.py`); deterministic, standard-library, self-checking.

Color is a subtraction. White light arrives; the molecule removes the photons whose energy matches an electronic gap, and what is left — transmitted or reflected — is the color we see. Four different gaps give the four families of molecular color, but the rule is one: a gap the size of a visible photon makes a colored substance.

## CC.0 The question

Why is copper sulfate blue, a carrot orange, cadmium-yellow paint yellow, and titanium-white paint white? This chapter derives the color of molecules and pigments from the electromagnetism of Chapter EM: visible light is the transverse electromagnetic wave (§EM.6), and **color is which visible photons a substance absorbs**, set by the size of an electronic energy gap.

## CC.1 The principle — color is selective absorption [F]

A substance has color when its electronic structure absorbs *part* of the visible spectrum (400–700 nm). An absorbed photon (a transverse light quantum) lifts an electron across an electronic **gap**; the gap size fixes which wavelength is absorbed, and the remaining light is the perceived color. For dissolved molecules the eye sees the **complementary** of the absorbed band; for opaque pigments it sees the **reflected** band. Two limits frame the rest: a gap *larger* than any visible photon absorbs nothing (white or colorless), and a gap *smaller* than every visible photon absorbs everything (black).

## CC.2 Energy and wavelength [F]

The bridge is the photon relation of Chapter EM,

$$E = \frac{hc}{\lambda}, \quad \lambda[\text{nm}] = \frac{1240}{E[\text{eV}]}.$$

Visible light spans  $E = 1.77$  eV (red, 700 nm) to 3.10 eV (violet, 400 nm). A gap in this energy window produces color; outside it, the substance is white/colorless (gap too large) or black (gap too small).

## CC.3 Mechanism 1 — $d$ – $d$ transitions (transition-metal complexes) [F]/[CAL]/[O]

In a transition-metal complex the ligands' electromagnetic  $1/r^2$  field splits the geometric  $d$ -orbitals by the crystal-field gap  $\Delta$  (Chapter CH §CH.5). An electron absorbs a photon to cross  $\Delta$ :

$$\lambda_{\text{abs}}[\text{nm}] = \frac{10^7}{\Delta[\text{cm}^{-1}]}.$$

For  $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$  ( $d^1$ , single transition),  $\Delta = 20300 \text{ cm}^{-1}$  gives  $\lambda_{\text{abs}} = 493 \text{ nm}$  (blue-green absorbed)  $\rightarrow$  reddish violet, matching observation (`vp_molecular_color.py`, §1). The mechanism — color *is* the  $d$ – $d$  gap — is forced by VP geometry [F]; the spectrochemical ordering of  $\Delta$  is calibration [CAL]; and for multi-electron ions ( $d^9 \text{ Cu}^{2+}$ ,  $d^8 \text{ Ni}^{2+}$ ) the single- $\Delta$  point model is insufficient (broad multi-band absorption tails into the red, giving the real blue/green), an honestly [O] limitation. Weak-field ligands give a small  $\Delta$  (red absorption); strong-field ligands a large  $\Delta$  (blue absorption).

## CC.4 Mechanism 2 — conjugated $\pi$ systems (organic dyes) [F]/[F?]

In a conjugated polyene the  $\pi$ -electrons behave as a free electron gas in a one-dimensional box of length  $L$  (the conjugation length). The levels are  $E_n = n^2 h^2 / (8m_e L^2)$ ; with  $N = 2k$  electrons ( $k$  double bonds) filling

levels up to  $k$ , the HOMO→LUMO gap is

$$\Delta E = \frac{h^2}{8m_e L^2} (N + 1).$$

Because  $L$  grows with conjugation, **the gap shrinks and the absorption red-shifts** — the derived trend (`vp_molecular_color.py`, §2): short polyenes (ethene, butadiene, hexatriene) absorb in the ultraviolet and are colorless, while extended conjugation (β-carotene, 11 double bonds) absorbs in the visible and is colored (orange; lycopene, red). The scaling  $\Delta E \propto (N + 1)/L^2$  and the UV→visible crossover are forced [F]; the absolute wavelengths from the free-electron model overestimate for long chains (bond alternation caps the gap), so the model’s numbers are trend-only [F?]. This is why carrots and tomatoes are colored: long  $\pi$ -conjugation pulls the gap down into the visible.

## CC.5 Mechanism 3 — the band gap (inorganic pigments and paint) [F]/[CAL]

A solid pigment is a semiconductor with a band gap  $E_g$ . Photons with  $E > E_g$  (wavelength below the absorption edge  $\lambda_{\text{edge}} = hc/E_g$ ) are absorbed; photons with  $E < E_g$  (longer wavelength) are reflected, and **the reflected band is the color**. As  $E_g$  decreases, the edge sweeps across the visible and the pigment runs **white → yellow → orange → red → black** (`vp_molecular_color.py`, §3):

| pigment                                             | $E_g$ (eV) | edge (nm) | reflected color               |
|-----------------------------------------------------|------------|-----------|-------------------------------|
| TiO <sub>2</sub> titanium white                     | 3.05       | 407       | white (all visible reflected) |
| ZnO zinc white                                      | 3.20       | 387       | white                         |
| As <sub>2</sub> S <sub>3</sub> orpiment             | 2.70       | 459       | yellow (violet/blue absorbed) |
| CdS cadmium yellow                                  | 2.42       | 512       | yellow                        |
| CdS · Se cadmium orange                             | 2.10       | 590       | orange/red                    |
| HgS vermilion (cinnabar)                            | 2.00       | 620       | red                           |
| Fe <sub>2</sub> O <sub>3</sub> red ochre (hematite) | 2.10       | 590       | red                           |
| CdSe cadmium red                                    | 1.73       | 717       | red (only deep red reflected) |
| carbon black                                        | ~0.5       | —         | black (all visible absorbed)  |

The same single rule  $\lambda_{\text{edge}} = hc/E_g$  orders the entire cadmium pigment series and the historical mineral pigments (vermilion, orpiment, hematite) [F], with  $E_g$  a measured material constant [CAL]. This is the physics of paint: **the band gap is the color knob**. Titanium white reflects everything because its gap (3.05 eV) sits just past the violet edge; cadmium red absorbs nearly all visible because its gap (1.73 eV) sits at the red edge.

## CC.6 Mechanism 4 — charge transfer (intense colors) [F]/[CAL]

When a photon moves an electron between a metal and a ligand (ligand-to-metal or metal-to-ligand charge transfer), the transition is fully allowed and therefore **intense** — 100–1000× stronger than a  $d$ – $d$  transition, so a trace gives a deep color. Permanganate  $\text{MnO}_4^-$  absorbs ~ 525 nm (O→Mn charge transfer) and is deep purple; dichromate  $\text{Cr}_2\text{O}_7^{2-}$  is orange; Prussian blue is an  $\text{Fe}^{2+} \leftrightarrow \text{Fe}^{3+}$  charge transfer (`vp_molecular_color.py`, §4). The mechanism is the same electron-across-a-gap; only the gap’s origin (metal↔ligand) and its allowedness differ.

## CC.7 White, black, and everything between [F]

The two achromatic limits follow from the same rule and require no extra assumption:

- **White:** gap larger than any visible photon ( $E_g > 3.1$  eV, edge in the ultraviolet). No visible light is absorbed; all is reflected. Titanium white, zinc white, table salt, snow.
- **Black:** gap smaller than every visible photon ( $E_g < 1.8$  eV, edge in the infrared). All visible light is absorbed. Carbon black, graphite.
- **Color:** gap inside the visible window. The substance absorbs the short-wavelength side of the edge and shows the long-wavelength side.

So white and black are not “colors added” but the two ends of where the gap sits relative to the visible band.

## CC.8 The unifying picture

All four families are one mechanism — **a transverse visible photon absorbed by lifting an electron across an electronic gap** — differing only in what sets the gap:

- *d*–*d*: the crystal-field splitting  $\Delta$  (ligand electromagnetic  $1/r^2$  field).
- conjugated  $\pi$ : the HOMO–LUMO gap  $\propto (N + 1)/L^2$  (conjugation length).
- band gap: the semiconductor  $E_g$  (the pigment’s electronic structure).
- charge transfer: the metal $\leftrightarrow$ ligand donor–acceptor gap.

The absorbing light is the transverse electromagnetic wave of Chapter EM ( $\chi \rightarrow 90^\circ$ ); only a photon whose energy matches the gap is absorbed (resonance), and the gap size determines the wavelength and hence the color. One principle — gap-sized absorption — and four ways nature builds the gap.

## CC.9 Falsification and reproducibility [F]

- If a transition-metal complex’s color did not track its crystal-field  $\Delta$  (weak-field red, strong-field blue absorption), the *d*–*d* account fails.
- If extending  $\pi$ -conjugation did not red-shift absorption (UV for short, visible for long), the particle-in-a-box account fails.
- If a semiconductor pigment’s color did not follow its band-gap edge  $\lambda_{\text{edge}} = hc/E_g$  (large gap white, small gap black, intermediate colored), the band-gap account fails — this is tested across the cadmium series and historical pigments.

```
python3 code/foundation/vp_molecular_color.py      # complementary color, d-d, FEM, band-
gap pigments
python3 code/foundation/vp_crystal_field.py        # crystal-field  $\Delta$  and 4/9
```

Both modules are deterministic (stdout sha256 identical on re-run) and standard-library only.

*Color completes the optical arc of Chapter EM: light emerges from the lattice, refraction and the rainbow split it by wavelength, and here matter subtracts\* selected wavelengths by lifting electrons across gaps — crystal-field, conjugation, band-gap, and charge-transfer gaps — so that a carrot is orange and titanium-white is white for the same reason, read through the size of a gap.\**

## Platinum Catalysis and the Electron Amplitude Wavelength: Toward Water Purification

**VP Chemistry & Electromagnetism Whitepaper — Chapter CT** DOI (reserved): 10.5281/zenodo.20680541 · Builds on Chapter EM, Chapter CH (§CH.5, §CH.11), and Chapter CM. Related to the VP Application whitepaper (10.5281/zenodo.18043066: Pt catalyst / electrolysis / desalination). Numbers reproduced by `vp_platinum_catalysis.py` (deterministic, standard-library). Grades distinguish **established** surface science ([CAL]/[F?]) from the **VP-framework hypothesis** ([H]).

A catalyst is a matchmaker: it must hold a reagent firmly enough to activate it, yet loosely enough to release the product. Platinum sits at that balance for hydrogen. This chapter asks whether the

balance can be read as a *resonance* of the electron’s amplitude wavelength with the metal surface — and whether that picture helps purify water.

## CT.0 The question

Why is platinum such a good catalyst, and can the “electron amplitude wavelength” of the VP framework illuminate the mechanism? The motivation is practical: platinum catalyses the hydrogen-evolution reaction in water electrolysis, which is central to producing clean hydrogen and to purifying water — the application this chapter targets.

## CT.1 Why platinum — states at the Fermi level [F?]

Platinum is  $[\text{Xe}]4f^{14}5d^96s^1$ : its  $5d$  shell has a single hole, so there are  $d$ -states right at the Fermi level. Electrons are therefore always available both to donate to an adsorbate and to accept back-donation from it. This is the prerequisite for surface catalysis, and it is exactly what gold ( $5d^{10}$ , filled) lacks — a filled  $d$ -shell makes gold comparatively inert, the same “filled versus unfilled  $d$ -shell” hinge that separated conductor from magnet in Chapter CM.

## CT.2 The Sabatier principle and the volcano [CAL]/[F?]

The Sabatier principle states that the best catalyst binds the key intermediate at *intermediate* strength: too weak and it cannot activate the reagent, too strong and it cannot release the product (it is poisoned). Plotting the hydrogen-evolution exchange current against the hydrogen adsorption free energy  $\Delta G_{\text{H}^*}$  gives a **volcano** peaking at  $\Delta G_{\text{H}^*} \approx 0$  (`vp_platinum_catalysis.py`, §1):

| metal     | $\Delta G_{\text{H}^*}$ (eV) | $\log_{10} j_0$ (A/cm <sup>2</sup> ) | regime                |
|-----------|------------------------------|--------------------------------------|-----------------------|
| <b>Pt</b> | <b>−0.09</b>                 | <b>−3.1</b>                          | <b>peak (optimal)</b> |
| Ir        | −0.10                        | −3.7                                 | good                  |
| Pd        | −0.20                        | −3.5                                 | good                  |
| Ni        | −0.28                        | −5.2                                 | good                  |
| Au        | +0.30                        | −5.4                                 | too weak              |
| W         | −0.55                        | −5.9                                 | too strong            |
| Hg        | +0.50                        | −12.6                                | too weak              |

Platinum has  $\Delta G_{\text{H}^*}$  closest to zero and the highest exchange current — it sits at the apex of the volcano. Tungsten and molybdenum bind hydrogen too strongly (cannot release it); gold and mercury too weakly (cannot activate it). The volcano data are calibration [CAL]; the Sabatier balance is the established organizing principle [F?].

## CT.3 The d-band center [CAL]/[F?]

The Hammer–Nørskov d-band-center model refines the picture: adsorption strength correlates with the mean energy  $\varepsilon_d$  of the  $d$ -band relative to the Fermi level. A higher  $\varepsilon_d$  (closer to the Fermi level) binds adsorbates more strongly. Platinum’s  $\varepsilon_d \approx -2.25$  eV lands in the optimal window, between the over-reactive early transition metals (high  $\varepsilon_d$ ) and the noble metals such as gold ( $\varepsilon_d \approx -3.56$  eV, too low, weak binding) (`vp_platinum_catalysis.py`, §2).

## CT.4 The electron amplitude wavelength — a VP resonance picture [H]

The VP framework adds a complementary, geometric reading. The catalytically active electron has a de Broglie *amplitude wavelength*

$$\lambda_e = \frac{h}{\sqrt{2m_e E}},$$

where  $E$  is the energy scale at which the electron engages the adsorbate — naturally the work function  $\varphi = 5.65$  eV for platinum. This gives  $\lambda_e = 5.16$  Å, so the **half-wavelength is**  $\lambda_e/2 = 2.58$  Å, commensurate with the platinum nearest-neighbour spacing  $d_{\text{Pt-Pt}} = 2.77$  Å — a matching ratio of **0.93** (`vp_platinum_catalysis.py`, §3).

The hypothesis: when the electron’s half-wavelength fits the surface geometry (one half-wave per surface bond), the active electron couples *resonantly* into the adsorbate’s antibonding orbital, so back-donation is enhanced and the adsorbate bond is weakened — the activation step of catalysis. Platinum’s particular combination of work function and lattice spacing places it inside this resonance window. This is a VP-framework picture, graded [H]: it reproduces the qualitative fact (platinum is exceptional) and supplies a geometric *why*, but it is a scale-matching argument, not a quantitative activity prediction. **§CT.7 puts it to a direct quantum simulation — which does not support the simple geometric form, and revises the picture accordingly.**

## CT.5 Tuning the resonance with potential — the electrolysis lever [H]

The amplitude-wavelength picture suggests a control knob. An applied electrode potential  $V$  shifts the effective energy of the active electron by  $eV$ , and therefore shifts  $\lambda_e$  (`vp_platinum_catalysis.py`, §4): a cathodic bias raises the electron energy and shortens the half-wavelength, an anodic bias lengthens it. Because catalysis happens *at* an electrode under applied potential, the electrode potential is precisely the lever that tunes the proposed resonance. At the thermodynamic water-splitting potential of 1.23 V the half-wavelength moves to 2.92 Å — a few percent off the lattice match — which the picture would read as a small detuning to be compensated by catalyst geometry. This makes the hypothesis *actionable* for electrode design rather than merely descriptive.

## CT.6 Application — water splitting and purification [F]/[CAL]

The target application is water. In electrolysis the platinum cathode runs the hydrogen-evolution reaction  $2\text{H}_2\text{O} + 2e^- \rightarrow \text{H}_2 + 2\text{OH}^-$  with an overpotential of only  $\sim 30$  mV — essentially zero — versus  $\sim 300$  mV for nickel (`vp_platinum_catalysis.py`, §5), so the cathodic energy loss is minimised. By Faraday’s law a current of 1 A for one hour produces 18.66 mmol ( $\sim 418$  mL at STP) of hydrogen ( $n = 2$ ).

For purification, electrolysis contributes three ways: (i) anodic oxidation breaks down microbes and organic contaminants (disinfection); (ii) the evolved hydrogen and reactive species reduce pollutants; and (iii) combined with the application whitepaper’s electromagnetic/electric-field desalination, it forms an integrated purification train. The thread back to this chapter: platinum’s catalytic efficiency — read in the VP picture as electron-amplitude-wavelength resonance — is what makes the low-voltage electrolysis (and hence the energy-efficient purification) feasible.

## CT.7 Simulation test of the resonance hypothesis [V]/refuted

Rather than rest on the scale coincidence, the hypothesis is put to a direct quantum simulation (`vp_pt_resonance_sim.py`): a 1D finite-difference Schrödinger model of a periodic platinum surface (wells spaced  $d = 2.77$  Å) with an adsorbate site, computing the adsorbate’s local density of states — the back-donation capacity — as a function of electron energy, hence wavelength, via the retarded Green’s function  $G = [(E + i\eta) - H]^{-1}$ .

The result **does not support the simple geometric resonance**, and the simulation says why:

1.  $\lambda_e = 2d$  is a **Bragg band edge, not a resonance**. A transfer-matrix band-structure calculation places the bottom of the first conduction band at  $\approx 4.5$  eV; the Bragg wavelength  $\lambda_e = 2d$  ( $E = 4.90$  eV) sits at that edge, with a gap below. The Bragg condition is where electrons are *reflected*, not resonantly transmitted.
2. **The adsorbate LDOS does not peak at  $\lambda_e = 2d$**  — there it is only  $\sim 2\%$  of its maximum. The simple “half-wavelength fits the lattice spacing  $\rightarrow$  enhanced back-donation” picture is therefore not borne out: geometric commensurability is not a dynamical resonance.
3. **What governs back-donation** is the metal density of states at the adsorbate energy together with the adsorbate’s own resonance — i.e. the established Newns–Anderson / d-band-center picture, not a geometric wavelength match. Platinum is exceptional because its partly filled  $5d$  band puts states at the Fermi level *and* its hydrogen binding sits at the Sabatier optimum, exactly the §CT.2–CT.3 account.
4. **Where a wavelength effect legitimately enters is quantum confinement**. Sweeping the slab thickness  $M$  (the confinement length  $L = Md$ ) at a fixed Fermi energy, the adsorbate LDOS *oscillates* strongly (coefficient of variation  $\approx 0.84$ , several reversals) — the real quantum-well-state effect known to modulate the reactivity of thin metal films. The wavelength that matters is matched to the film *thickness*, not to the in-plane Pt–Pt spacing.

So the simulation refines the chapter rather than confirming it: the  $\lambda_e/2 \approx d_{\text{Pt–Pt}}$  coincidence of §CT.4 is a coincidence, not a back-donation resonance.

## CT.8 Is reproduction impossible? — energy resonance reproduces platinum [V]

The right question, having refuted the geometric form, is *whether platinum’s catalysis can be reproduced at all, or whether the failure is fundamental*. It is not fundamental: the failure was a **wrong choice of resonance variable**, and the correct one reproduces platinum (`vp_pt_reproduction.py`).

The catalytic resonance is in **energy**, not geometry: the adsorbate orbital  $\varepsilon_a$  resonates with the metal’s **d-band centre**  $\varepsilon_d$  (the Newns–Anderson picture). A Newns–Anderson simulation with a semi-elliptical d-band self-energy  $\Sigma = V^2 g_d$  shows the sharp adsorbate level broadening into a resonance on the d-band — the energy-resonance mechanism. Crucially, the **d-band centre predicts the hydrogen binding**: across Ni, Rh, Pd, Ir, Pt, Cu, Au, the (independently determined)  $\varepsilon_d$  correlates with the measured  $\Delta G_{\text{H}^+}$  at  $r = -0.91$  (a non-circular test, since  $\varepsilon_d$  comes from band structure, not from the binding). Reconstructing the Sabatier volcano (activity  $\propto \exp(-|\Delta G_{\text{H}^+}|/kT)$ ) then places **platinum (with iridium) at the apex** — the experimental result, reproduced from the electronic structure.

So the answer to “why couldn’t we reproduce it, and is it impossible?” is:

- **The geometric resonance failed because it is genuinely wrong** —  $\lambda_e = 2d$  is a Bragg band edge in any dimension and any model (robust solid-state physics), not a limitation of the 1D simulation.
- **Platinum’s catalysis is fully reproducible** — but through the *energy* resonance ( $\varepsilon_a \leftrightarrow \varepsilon_d$ ), which reproduces the binding ( $r = -0.91$ ) and the volcano apex (Pt).
- **The “electron amplitude wavelength” must be reinterpreted**: not the de Broglie wavelength matched to the lattice constant, but the **d-electron energy** matched to the adsorbate orbital. Read that way, the resonance intuition is correct and platinum is reproduced.

The catalytic essence of platinum is therefore the **energy** of the d-states that its one-hole-short  $5d$  shell places at the Fermi level — exactly the §CT.1–CT.3 account, now shown to be a resonance, just an energetic one.

## CT.9 Assessment of the possibility [H]→reframed

Honestly weighed across both simulations:

- **The resonance intuition was right; the variable was wrong**. Catalysis *is* a resonance — of the adsorbate orbital with the d-band (energy), not of the electron wavelength with the lattice (geometry). The energy version reproduces platinum at the volcano apex ( $r = -0.91$ ); the geometric version is a band edge.

- **What survives of the amplitude-wavelength idea.** Two things: (i) the correct reading of “amplitude wavelength” as d-electron *energy*, which is the established and predictive d-band-centre descriptor; and (ii) a genuine, simulable *quantum-confinement* effect (thin-film quantum-well states, §CT.7) by which electron wavelength modulates reactivity through film thickness — an actionable lever for the water-splitting electrodes of §CT.6.
- **Verdict.** Reproduction is not impossible; it is achieved here through the energy resonance. The honest, predictive descriptor is the d-band centre, and the constructive engineering levers are the d-band position (alloying, strain) and quantum confinement (film thickness, potential) — both tied directly to designing efficient catalyst layers for water electrolysis and purification.

This is the value of pressing past “refuted” to “why, and is it possible”: the geometric coincidence is discarded, the energy resonance is shown to reproduce platinum, and the amplitude-wavelength idea is corrected into its working form rather than abandoned.

## CT.10 Falsification and reproducibility [F]/[V]

- Established: if the hydrogen-evolution volcano did not peak near  $\Delta G_{H^*} = 0$ , or if platinum were not near that peak, the Sabatier account fails.
- Hypothesis (now tested): the §CT.4 amplitude-wavelength resonance predicted that the adsorbate back-donation capacity would peak when  $\lambda_e/2$  matches the Pt–Pt spacing. The simulation of §CT.7 finds the opposite —  $\lambda_e = 2d$  is a band edge and the adsorbate LDOS does not peak there — so this specific form is **refuted**, and the surviving, testable claim is the quantum- confinement one: thin-film reactivity should oscillate with layer thickness (and with  $E_F$  under applied potential).

```
python3 code/foundation/vp_platinum_catalysis.py # volcano, d-band,  $\lambda_e$  scale-matching, electrolysis
python3 code/foundation/vp_pt_resonance_sim.py   # geometric resonance test: band edge (refuted)
python3 code/foundation/vp_pt_reproduction.py     # energy resonance: reproduces Pt at volcano apex
```

All three modules are deterministic (stdout sha256 identical on re-run) and standard-library only.

*Platinum extends the metals thread of Chapter CM — filled d gives conduction (copper), unfilled d gives magnetism (iron), and a d-shell one hole short of full gives catalysis (platinum). The VP reading proposed a resonance of the electron’s amplitude wavelength with the surface. Put to a quantum simulation, the geometric\* form fails (the Bragg match is a band edge), but pressing on to ask whether reproduction is possible shows that it is: the energy resonance — the adsorbate orbital with the d-band centre — reproduces platinum at the volcano apex ( $r = -0.91$ ). The resonance intuition was right; the variable was energy, not geometry. What this buys for water electrolysis and purification is concrete: the levers are the d-band position and quantum confinement of thin catalyst layers, both tuning the Fermi-level d-states whose energy is the true seat of platinum’s catalysis.\**

## Applications: Catalysis, Fertilizer, Water, Energy, and Materials

**VP Chemistry & Electromagnetism Whitepaper — Chapter CA** DOI (reserved): 10.5281/zenodo.20680541 · Companion to Chapters CH (chemistry), CM (metals), CT (catalysis), EM (electromagnetism). Status grades: [F] forced (zero free parameters) · [F?] forced-candidate · [CAL] calibration input (measured constant) · [V] simulation-measured · [H] hypothesis · [O] open / refuted. Every numeric claim is reproduced by a deterministic, standard-library module (2× sha256 identical); module and case-ledger anchors are given inline.

This chapter carries the metals account (CM/CT) into five practical domains: **catalysts, fertilizer, water purification, energy storage, and new materials**. Its method is the one that already succeeded for platinum (Chapter CT): where a speculative *amplitude/geometry* mechanism was proposed, it is tested; what does not survive is reported as refuted and **re-grounded on validated physics**; what does survive is kept and labelled. The result is a single thread — the

**d-band energy resonance** — running through catalysis, fertilizer, and water electrolysis, with honest negative verdicts for two device claims that do not hold.

## CA.0 The central method: re-grounding, not decoration

The prior application materials built catalysis, desalination, and an energy device on an *electron amplitude* variable (fm-scale). Chapter CT showed by direct simulation that this variable is the wrong one: the geometric/amplitude resonance is a Bragg band edge (refuted), and platinum is reproduced only by the **energy** resonance — the d-band center ( $\varepsilon_d$ ) coupling to the adsorbate level (Newns–Anderson / Hammer–Nørskov). This chapter applies that verdict everywhere: each application is rebuilt on the validated mechanism, and any device claim whose mechanism fails testing is reported as **refuted** or **[H]** with a working alternative supplied. Honesty about the two negative results (magnetic desalination; the ESS “amplitude  $\rightarrow$  electricity” generator) is not a footnote — it is the chapter’s spine.

## CA.1 The d-band engine — one variable for three reactions [V]/[F?]

Module `vp_dband_catalysis.py` generalises the platinum result into a reusable engine. A sharp adsorbate level broadens on coupling to the metal d-band (Newns–Anderson) [V]; the d-band center  $\varepsilon_d$  then *predicts* binding. For hydrogen the correlation with measured adsorption is

$$r(\varepsilon_d, \Delta G_H) = -0.915 \quad (\text{strong, non-circular}) \text{ [F?]},$$

and the Sabatier volcano (activity peaked at  $\Delta G_H \approx 0$ ) reproduces the **Pt apex**. The *same engine* applied to nitrogen adsorption gives a different optimum and the **Fe/Ru apex** of the ammonia volcano. Crucially, the real tuning lever is the **energy**, not the lattice spacing: shifting  $\varepsilon_d$  by alloying or strain moves a metal along the volcano (`vp_dband_catalysis.py` §4). This *replaces* the refuted geometric resonance with a design rule that is actually used for electrodes. **Falsifiable:** HER apex at  $\Delta G_H \approx 0$  (Pt), ammonia apex at  $E_N \approx -0.5$  eV (Fe/Ru); break either ordering and the d-band picture is discarded.

## CA.2 Fertilizer — ammonia synthesis [F]/[F?]

Nitrogen fixation  $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$  underpins half the human food supply. Module `vp_ammonia_synthesis.py` closes it with the two VP pillars — equilibrium (CH.8) and the d-band catalyst engine (CA.1).

**Equilibrium forces the conditions [F].** With  $\Delta H^\circ = -91.8$  kJ/mol (exothermic) and  $\Delta S^\circ = -198.1$  J/mol  $\cdot$  K (gas moles  $4 \rightarrow 2$ ),  $\Delta G^\circ = \Delta H - T\Delta S$  gives the crossover  $T^* = \Delta H/\Delta S = 463$  K, where  $K = 1.000$ . Below it the reaction is favoured but  $\text{N}_2$  is kinetically inert; above it the equilibrium yield collapses. The module solves the stoichiometric equilibrium and reproduces the **Le Chatelier compromise** quantitatively — e.g. equilibrium  $\text{NH}_3$  rises with pressure ( $\Delta n = -2$ ) and falls with temperature:

| condition | 1 atm | 100 atm | 200 atm | 400 atm |
|-----------|-------|---------|---------|---------|
| 500 K     | 10.8% | 76.3%   | 82.6%   | 87.3%   |
| 700 K     | 0.6%  | 29.1%   | 40.7%   | 52.4%   |

This is exactly why industry runs  $\approx 450^\circ\text{C}$  / 150–300 atm / Fe catalyst — a forced three-way compromise, not a tuned choice.

**Catalysis selects the metal [F?].** The rate-limiting step is dissociation of  $\text{N} \equiv \text{N}$  (945 kJ/mol, among the strongest bonds; the bond-energy sum reproduces  $\Delta H \approx -93$  kJ/mol, sign and magnitude correct). The catalyst volcano in the nitrogen-binding descriptor peaks at **Fe/Ru/Os** — the industrial Haber–Bosch catalysts — with Mo/W too strong (N trapped) and Ni/Pt too weak (no dissociation). The volcano comes from the d-band energy of CA.1, one thread with catalysis and water-splitting. Absolute rates, promoter ( $\text{K}_2\text{O}/\text{Al}_2\text{O}_3$ ) effects, and surface microkinetics remain [O].

### CA.3 Water purification I — electrolysis electrodes [F]/[F?]

Module `vp_water_electrolysis.py` carries the d-band engine to clean hydrogen and electrochemical water treatment. Water splitting  $2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$  has a reversible minimum  $E_{\text{cell}}^\circ = 1.229 \text{ V}$  ( $\Delta G = 237 \text{ kJ/mol H}_2$ ) and a thermoneutral voltage  $V_{tn} = \Delta H/(nF) = 1.481 \text{ V}$ ; the difference  $T\Delta S = 48.6 \text{ kJ/mol}$  is supplied as heat [F]. The real cell voltage decomposes as  $V_{\text{cell}} = 1.23 + \eta_{\text{HER}} + \eta_{\text{OER}} + iR$ , and the single largest loss is the **anode (OER)**. The HER volcano (CA.1) places the cathode near solved (Pt apex). The OER is bounded by a *fundamental* limit: the universal scaling relation  $\Delta G_{*\text{OOH}} - \Delta G_{*\text{OH}} \approx 3.2 \text{ eV}$  ties the intermediates, so the two middle steps cannot both be 1.23 eV and the best achievable overpotential is

$$\eta_{\text{OER}}^{\min} = 1.6 - 1.23 = 0.37 \text{ V} \quad [\text{F?}],$$

with the apex at **RuO<sub>2</sub>/IrO<sub>2</sub>** (the industrial OER catalysts). This 0.37 V is *why water-splitting is permanently expensive* on single-site catalysts — beating it requires breaking the scaling relation (bifunctional / 3D sites), an honest [O]. Faraday’s law gives 18.66 mmol H<sub>2</sub> per A · h [F] and a thermoneutral voltage efficiency  $V_{tn}/V_{\text{cell}} \approx 81\%$ . Electrocoagulation, electrodialysis, and capacitive deionization (CDI) extend the same electrode science to treatment and desalination — all charge/membrane/potential based, [F?]/[CAL].

### CA.4 Water purification II — magnetic desalination: an honest verdict [F] · [O]/refuted

The prior materials proposed separating H<sub>2</sub>O from Na<sup>+</sup>/Cl<sup>−</sup> at 0.45 T by an *amplitude* difference. Module `vp_magnet_desalination.py` replaces the prose with the actual magnetic force. Water is weakly diamagnetic ( $\chi_v \approx -9.05 \times 10^{-6}$ ); its magnetic energy in a 0.45 T field is

$$\frac{U_{\text{mag}}}{k_B T} \approx 5.3 \times 10^{-9} \quad [\text{F}],$$

nine orders of magnitude below thermal energy — thermal motion overwhelms any magnetic ordering, so **no separation occurs**. To reach  $U_{\text{mag}} \approx k_B T$  would need  $B \approx 6173 \text{ T}$ , physically impossible for a purifier (cf. ~45 T continuous, ~1200 T pulsed-destructive). The Lorentz force on flowing ions is real but  $\sim 10^{-2} k_B T$  across a channel: a field can stir (MHD) but cannot *desalinate by amplitude*. **Verdict: the amplitude-separation device is refuted [O]**. The working alternatives are supplied with real numbers: reverse osmosis (seawater osmotic pressure  $\pi = i c R T \approx 30 \text{ bar}$ , matching the measured ~27 bar; thermodynamic minimum  $\approx 0.83 \text{ kWh/m}^3$ ), electrodialysis, and CDI. This is the platinum pattern again — refute the wrong mechanism, keep the validated one.

### CA.5 Energy storage — black-copper solar ESS, split honestly [F]/[F?] · [H]→alt

Module `vp_ess_thermal.py` separates the device into a **real storage part** and a **hypothetical generation part**.

**Storage is real and quantified.** A black-copper absorber ( $\alpha \geq 0.98$ ) at 800 W/m<sup>2</sup> collects ~4.7 kWh/day per m<sup>2</sup>. High-heat-capacity media (alumina/sand) store it; the heat capacity is the Dulong–Petit  $3R$  per atom of Chapter CH.7 ( $c_p \approx 1223 \text{ J/kg} \cdot \text{K}$  at the high-T limit; measured 880 J/kg · K at room temperature), so a 1 m<sup>3</sup> store holds **~46 kWh** of heat over a 118 K rise [F?]. Vacuum insulation gives a thermal time constant  $\tau = mc/(UA) \approx 5.4 \text{ days}$  [F?] — ample for overnight delivery.

**Generation is hypothesis, with a working alternative.** The claimed “rotation-amplitude → electric field” generator has no validated mechanism and is marked [H]. Heat-to-electricity is bounded by Carnot,  $\eta = 1 - T_c/T_h = 28.2\%$  at the storage temperature; realistic conversion uses a thermoelectric (Seebeck,  $ZT = 1 \rightarrow \sim 5.5\%$ , ~2.5 kWh) or a Stirling engine (~½ Carnot → ~14%, ~6.5 kWh) [F]. The honest split — real thermodynamic storage, unproven generator replaced by a heat engine — is the same discipline as CA.4.

## CA.6 New materials — black copper and d-band design [F]/[F?]

Module `vp_new_materials.py` explains the black-copper absorber by *optics*, not amplitude, and ties materials design to the d-band. Polished copper reflects ( $R \approx 0.97$ , absorptance  $\alpha = 0.03$ ) because of its free-electron plasma (Chapter CM); nanostructured/oxidised copper raises  $\alpha$  to  $\sim 0.96$  (a  $\sim 32\times$  gain) by graded-index impedance matching, light-trapping, and intrinsic CuO absorption [F?] — the same electromagnetic physics as the copper plasma/skin effect, not an fm-scale amplitude. A **selective absorber** then needs *low* infrared emittance: the Stefan–Boltzmann re-radiation  $P = \varepsilon\sigma A(T^4 - T_{\text{amb}}^4)$  at the storage temperature is  $+704$  W net for  $\varepsilon = 0.05$  but  **$-377$  W for a blackbody surface** ( $\varepsilon = 0.90$ ) — the latter *cannot even heat up* [F]. Thus low IR emittance is essential, a concrete falsifiable design rule. Beyond coatings, the d-band of CA.1 is the materials lever (alloying/strain tunes catalytic, electronic, and magnetic properties — the CM/CT trilogy), and crystal packing fractions remain pure geometry (FCC/HCP  $\pi/(3\sqrt{2}) = 0.7405$ , [F], CH.13).

## CA.7 Master ledger — applications

[F] **forced (zero free parameters)**: ammonia  $T^* = \Delta H/\Delta S$ , Le Chatelier directions, bond-energy  $\Delta H$  sign; water-splitting 1.23/1.48 V and Faraday 18.66 mmol/A · h; magnetic energy  $\ll k_B T$  and the infeasible  $B$ ; seawater osmotic pressure and desalination minimum energy; Dulong–Petit storage, thermal time constant, Carnot/Seebeck/Stirling efficiencies; Stefan–Boltzmann selective-absorber balance; crystal packing fractions.

[F?] **forced-candidate**:  $\varepsilon_d \leftrightarrow$  binding correlations and volcano apexes (HER→Pt,  $\text{NH}_3 \rightarrow \text{Fe/Ru}$ , OER→ $\text{RuO}_2/\text{IrO}_2$ ); d-band strain/alloy tuning; OER scaling-relation limit 0.37 V; absorber enhancement and selectivity.

[V] **simulation-measured**: Newns–Anderson adsorbate-DOS broadening; volcano reconstructions.

[CAL] **calibration inputs**:  $\varepsilon_d$ ,  $\Delta G_H$ ,  $E_N$ , OER descriptors,  $\Delta H/\Delta S$ , bond energies,  $E^\circ$ , susceptibility, emittance, insolation.

[H] **hypothesis** → **alternative**: the ESS “amplitude → electricity” generator (replaced by a heat engine).

[O] **open / refuted**: magnetic amplitude-desalination (refuted); absolute reaction rates and microkinetics; promoter effects; scaling-relation circumvention.

## CA.8 Falsification criteria [F]

- If the HER apex is not Pt ( $\Delta G_H \approx 0$ ) or the ammonia apex is not Fe/Ru ( $E_N \approx -0.5$  eV), the d-band volcano account is discarded.
- If exothermic ammonia equilibrium does not fall with temperature and rise with pressure, the equilibrium account is discarded.
- If a single-site OER catalyst stably shows  $\eta < 0.3$  V, the scaling-relation limit needs revision.
- If a 0.45 T field measurably desalinates seawater, the magnetic-energy analysis is wrong (it is not:  $U_{\text{mag}}/k_B T \sim 10^{-9}$ ).
- If a non-selective (high- $\varepsilon$ ) black absorber maintains the storage temperature under the stated flux, the radiative-balance rule is wrong.

## CA.9 Reproducibility

Six deterministic, standard-library modules and three case ledgers (29 rows) reproduce this chapter; `verify_applications.py` checks execution, determinism ( $2\times \text{sha256}$ ), standard-library-only dependency, and ledger integrity in one pass (**6/6 PASS, 29 rows**).

```
python3 verify_applications.py      # all modules + ledgers
python3 vp_dband_catalysis.py      # HER→Pt, NH3→Fe/Ru, d-band tuning lever
python3 vp_ammonia_synthesis.py    # Haber compromise, T*, catalyst volcano
python3 vp_water_electrolysis.py   # 1.23/1.48 V, OER 0.37 V limit, Faraday
```

```
python3 vp_magnet_desalination.py    # magnetic separation refuted; RO/ED/CDI
python3 vp_ess_thermal.py            # thermal storage (CH.7); generation [H] + heat engine
python3 vp_new_materials.py          # black-copper optics; selective absorber; packing
```

*The five applications close on one idea carried from the metals trilogy: the **d-electron energy** (not a lattice geometry, not an amplitude) governs catalysis, and the same energy thread reaches fertilizer and water-splitting; thermodynamics governs equilibrium and storage; optics governs the absorber. Where a device mechanism could not be validated it is refuted or marked hypothesis and a working alternative is supplied. What is forced is falsifiable, what is measured is labelled, what is open is admitted — and all of it is reproducible.*

## Application: CO<sub>2</sub> Electrocatalytic Reduction — the *d*-band CO-binding descriptor [F]/[F?]

Numbers reproduced by `vp_co2_reduction.py` (deterministic, standard-library,  $2\times \text{sha256}$ ). This section re-grounds the carbon-dioxide problem on the verified *d*-band energy descriptor, replacing the refuted electron-amplitude account (in which CO<sub>2</sub> “dissociated” when an amplitude exceeded  $\sim 300 \text{ fm} > \sqrt{2}$ ). The amplitude figure had no measured basis; the controlling variable is the \*CO adsorption energy, exactly as for the hydrogen and nitrogen volcanoes (Chapter CT, §CA).

### The mechanism — selectivity is set by \*CO binding [F?]

Electrochemical CO<sub>2</sub> reduction (CO<sub>2</sub>RR) proceeds through adsorbed intermediates, and its *product selectivity* is governed by a single descriptor: the binding energy of adsorbed CO,  $\Delta E_{\text{CO}}$  — the same *d*-band quantity that orders hydrogen ( $\Delta G_H$ , apex Pt) and nitrogen ( $E_N$ , apex Fe/Ru). The rate-limiting first electron transfer to \*CO<sub>2</sub><sup>−</sup> / \*COOH / \*OCHO is what a catalyst must stabilize; how well it does so tracks  $\Delta E_{\text{CO}}$ . The module finds  $\Delta E_{\text{CO}}$  correlated with the Hammer–Nørskov *d*-band centre  $\varepsilon_d$  at  $r = -0.97$  — confirming the descriptor and reusing the same engine, not a new one.

### Thermodynamics — equilibria near 0 V, onset far below [F]

The multi-electron product potentials (vs RHE) cluster near zero, yet the observed onset is much more negative (−0.8 to −1.1 V) because the CO<sub>2</sub><sup>•−</sup> radical anion lies near −1.9 V:

| Product                       | $n(e^-)$ | $E^\circ$ (V vs RHE) |
|-------------------------------|----------|----------------------|
| CO                            | 2        | −0.10                |
| HCOOH (formate)               | 2        | −0.12                |
| CH <sub>3</sub> OH            | 6        | +0.03                |
| C <sub>2</sub> H <sub>4</sub> | 12       | +0.08                |
| CH <sub>4</sub>               | 8        | +0.17                |
| (competing) H <sub>2</sub>    | 2        | 0.00                 |

These equilibria are forced from formation free energies by  $E^\circ = -\Delta G^\circ/(nF)$ ; the numeric gate independently reproduces  $E^\circ(\text{CO}_2 \rightarrow \text{CO}) = -0.104 \text{ V}$  and  $E^\circ(\text{CO}_2 \rightarrow \text{CH}_4) = +0.169 \text{ V}$  from  $\Delta G_f^\circ$  data.

### Copper’s uniqueness — the falsifiable headline [F?]

Sorting metals by  $\Delta E_{\text{CO}}$  reproduces Hori’s four-group classification:

- **Strong binding** (Pt, Pd, Ni, Rh, Co, Ru): the surface is CO-poisoned and evolves H<sub>2</sub>.
- **Intermediate binding** ( $\Delta E_{\text{CO}} \approx -0.5 \text{ eV}$ ): CO stays long enough to be reduced further to **hydrocarbons and alcohols** — this is **copper, and copper alone** among pure metals.
- **Weak binding** (Au, Ag, Zn): CO desorbs and is the product (CO-selective).
- ***sp*-metals** (Sn, Pb, Bi, In, Cd): bind \*OCHO through oxygen and give **formate**.

That copper is the only pure metal producing significant C<sub>2</sub>+ hydrocarbons is the central, established experimental fact (Hori), reproduced here as a consequence of intermediate \*CO binding — the CO<sub>2</sub>RR analogue of “Pt is the HER apex.”

### The scaling-relation ceiling [F?]

Because \*COOH, \*CO, and \*CHO all bind through carbon, their energies scale together with a nearly constant offset  $\Delta E_{\text{CHO}} \approx \Delta E_{\text{CO}} + 0.74 \text{ eV}$ . The \*CO → \*CHO step is therefore potential-limiting, giving a methane

limiting potential on copper of  $U_L \approx -0.74$  V (Peterson–Nørskov). This 0.74 eV floor cannot be removed by tuning  $\varepsilon_d$  alone — a linear-scaling constraint that is the direct cousin of the OER 0.37 V overpotential floor in the water chapter. It is the reason CO<sub>2</sub>RR is intrinsically difficult, stated as forced geometry of the scaling lines, not as a fitted loss.

## Grade and falsification

**Forced [F]:** the equilibrium potentials and electron counts. **Candidate-forced [F?]:** the  $\varepsilon_d \leftrightarrow \Delta E_{\text{CO}}$  descriptor, copper’s uniqueness, the volcano, and the scaling ceiling. **Calibration [CAL]:**  $\Delta E_{\text{CO}}$ ,  $\varepsilon_d$ , the equilibrium potentials, and the scaling offset. **Open [O]:** absolute current densities and the full C–C coupling microkinetics. **Falsifier:** copper is the apex for hydrocarbons and CO<sub>2</sub>RR selectivity follows the \*CO-binding order; if a pure non-copper metal produces more hydrocarbons at the same overpotential, or the binding order is broken, the  $d$ -band CO descriptor is refuted.

## Appendix O: Open-Items Register (what is honestly unsolved)

This appendix consolidates every [O] / [H] item carried in this volume, with the specific physics that would be required to close it. None is a defect to be fixed; each is open research. Closing any of them would be *new physics*, not a correction — and forcing a closure would reintroduce exactly the kind of overfitting (the retracted  $\sqrt{2.5}$  rule, the refuted electron amplitude) that this program has been careful to discard.

### O.1 Chemistry — peripheral quantitative opens (independently re-examined)

Each was re-examined by `vp_open_items_reexam.py`; all four correctly remain open because the closing variable lies *outside* VP  $\pi$ -geometry.

| Open item                                 | Module                              | Closing physics (outside VP geometry)                                                                                                                                                          |
|-------------------------------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Absolute bond-dissociation energy         | <code>vp_bond_energy.py</code>      | the Morse range $\beta = a r_e$ (anharmonicity; lone-pair / polarity). A direct test found $\beta$ uncorrelated with bond order ( $r = +0.18$ ), so $\sqrt{2}$ geometry alone cannot close it. |
| Defect knockdown (real vs ideal strength) | <code>vp_strength.py</code>         | dislocation density $\rho_{\text{disl}}$ (processing-dependent; Taylor $\tau = \alpha G b \sqrt{\rho}$ ) — intrinsically a calibration input.                                                  |
| Naïve shell correction                    | <code>vp_shell_correction.py</code> | the nuclear spin-orbit shell model (Mayer–Jensen); a uniform proximity bonus over-corrects because $\Delta B$ changes sign across magic numbers.                                               |
| Electron affinity / absolute IE           | <code>vp_dblock_chemistry.py</code> | electron screening constants (Clementi–Raimondi); Slater rules give only the correlation (sign correct 8/18).                                                                                  |

### O.2 Electromagnetism — foundational opens (deeper than the chemistry set)

| Open item                                                                        | Grade | Status                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Electromagnetic coupling <math>\alpha_{\text{em}}</math></b><br>(the “1/137”) | [CAL] | The <i>strength</i> of electromagnetism is a measured input; the framework derives the <i>form</i> ( $1/r^2$ , $c$ , $E/B$ , polarization, the angle law) but not the magnitude. Physics §14.5 states it is not derived. A topological / phase-winding route has been scaffolded but is computationally heavy and is <b>on hold</b> ; there is no known low-cost shortcut. Closing it is new physics. |

| Open item                                             | Grade | Status                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Long-range / unscreened EM</b><br>(global $U(1)$ ) | [H]   | Masslessness and infinite range rest on the breaking of a <i>global</i> (not gauged) $U(1)$ , giving a Goldstone mode — a stated departure from gauge theory. The AQD dynamics supply causality (§EM.12) but not the gauge structure; this remains a hypothesis.                                                                                                                                                                                                                                                      |
| <b>Absolute vector (magnetic / radiative) sector</b>  | [O]   | The longitudinal (Coulomb / $E$ ) sector — including the dynamic equations, Poynting flow, and causality — is closed in §EM.12. The full vector $\mathbf{E}/\mathbf{B}$ curl structure (Faraday's $\nabla \times \mathbf{E}$ and Ampère–Maxwell's $\nabla \times \mathbf{B}$ in vector form, with the magnetic field as an independent transverse pseudovector and its absolute magnitude) is <b>not</b> supplied by the scalar/longitudinal treatment. This is the principal open structural item of the EM chapter. |
| $\gamma\gamma$ genesis dynamics                       | [O]   | How charge is created from photon collisions ( $\gamma\gamma \rightarrow e^+e^-$ ) at the level of lattice dynamics is open.                                                                                                                                                                                                                                                                                                                                                                                          |

### O.3 What is *closed* (for contrast)

The longitudinal electrodynamics is complete and reproducible: light emergence ( $c^2 = B/\rho$ ), the static Coulomb  $1/r^2$ , the  $|\mathbf{B}| = (v/c)|\mathbf{E}|$  ratio and  $\nabla \cdot \mathbf{B} = 0$ , polarization as helicity, the conduction–radiation angle, blackbody radiation, refraction, and — as of this edition — the **dynamic field equations, Poynting's theorem, energy conservation, uniqueness, and light-cone causality** (§EM.12). On the chemistry side the forced body is complete: bonding geometry, coordination color, the metal  $d$ -shell fates, thermodynamics, equilibrium, kinetics, solution and electrochemistry, materials geometry, and the nuclear skeleton; and the applications are re-grounded on the verified  $d$ -band descriptor (catalysis, ammonia, water electrolysis, energy storage, new materials,  $\text{CO}_2$  reduction), with the non-working mechanisms (magnetic desalination, amplitude-to-electricity ESS) honestly refuted and replaced by working alternatives.

## Conclusion

This volume completes a single arc from one anchor. **Light emerges** from the jammed lattice as the unique surviving longitudinal wave at  $c^2 = B/\rho$ , shown in closed form and by a deterministic simulation (0.06%), which also recovers linear dispersion, the collapse of the transverse branch, and the framework's  $-19$  ppm mass-ratio residual (§EM.2). On that foundation **charge is rotation**, the Coulomb  $1/r^2$  is three-dimensional flux conservation,  $E$  and  $B$  are the longitudinal and transverse parts of one twist, polarization is helicity, and one wave spans conduction and radiation by a single angle (§EM.3–EM.8). This edition closes the **dynamic and causal structure** of that wave in longitudinal form — the first-order field equations, the Poynting energy flow with exact conservation, uniqueness, and a strict light cone with quantum microcausality (§EM.12, from the AQD axiomatic dynamics).

**Chemistry follows by sphere geometry and rotation.** Points *on* a sphere give molecular shape ( $\arccos(-1/3)$ ); spherical harmonics under a ligand field give the crystal-field ratio  $4/9$ ; sphere *packing* gives  $0.7405$ ; lattice points *inside* a sphere give the nuclear magic skeleton. Rotation threads the rest — the heat-capacity staircase,  $\Delta G = \Delta H - T\Delta S$ , the Boltzmann tail, and rotational-dent transfer (the Daniell cell at  $1.10$  V). The metal chapters split the  $d$ -shell into conduction (copper) and magnetism (iron) and into color ( $d$ - $d$  transitions); the catalysis chapter records, in full, how the early electron-amplitude resonance picture was **refuted** (a Bragg band edge, not a resonance) and **reproduced correctly** by the  $d$ -band energy descriptor.

**The applications close the arc on verified physics.** A single  $d$ -band engine explains hydrogen (apex Pt), nitrogen for ammonia (apex Fe/Ru), and **CO<sub>2</sub> reduction (the uniqueness of copper for hydrocarbons)**; water electrolysis is bounded by the OER scaling floor; energy storage is real in its Dulong–Petit heat capacity and Carnot-bounded in its conversion; and the mechanisms that do *not* work — magnetic desalination, amplitude-to-electricity generation — are refuted honestly and replaced by working alternatives. The same discipline that discards a beloved but unworkable idea is what lets the rest be built upon.

What is **forced** is falsifiable and stated with its criterion; what is **measured** is labelled and never tuned per result; what is **open** — most fundamentally the electromagnetic coupling  $\alpha_{\text{em}}$ , and the absolute vector sector of electromagnetism — is admitted in the Open-Items Register (Appendix O) rather than disguised. The result is intended to be both substantially more detailed than the original working notes and fully reproducible: a 38-module chemistry/EM core with a 176-entry case ledger and a 9-module / 46-entry applications package, verification harnesses (38/38 and 9/9), and numeric-drift gates (both pass, zero drift) accompany this text, and every number above can be regenerated by the reader.

## Reproduction (summary)

```
# chemistry + EM core
python3 code/verify_chemistry.py --root .      # 38/38 modules pass; ledgers 114 + 62 pass
python3 code/gate_numeric_chem.py              # numeric drift gate: PASS (zero drift)
python3 code/foundation/vp_light_emergence.py  # light emerges at  $c^2 = B/\rho$ 
python3 code/foundation/vp_em_field_dynamics.py # dynamic field, Poynting, causality (EM.12)

# applications
python3 apps/verify_applications.py            # 9/9 modules pass; ledger 46 rows
python3 apps/vp_chem_numeric_ssot.py          # applications drift gate: PASS (17 anchors)
python3 apps/vp_co2_reduction.py              # CO2 reduction: copper uniqueness

sha256sum -c SHA256SUMS.txt                   # file integrity
```

## Data and code availability

All code, data, and documents are released under open licenses (documentation and data CC BY 4.0; code MIT) in the DOI package reserved as 10.5281/zenodo.20680541, with checksums and a deterministic RunID.

The package is a supplement to the VP Theory whitepaper (10.5281/zenodo.17932567), takes its field-dynamics addendum from the AQD record (10.5281/zenodo.17423870), and supersedes the earlier qualitative application notes (related: 10.5281/zenodo.18043066).